



BobSim

FourPostEval Suspension Kinematics and Compliance

Heave and roll sweeps

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Run Notes

- Heave and roll sweeps evaluate camber, toe, caster, KPI, trail, scrub, motion ratio, and jacking characteristics.
- Spring and stabar rates are applied from `model_overrides.suspension` before the model run.
- Summary metrics use the same KnC-style table layout as the legacy report.

FourPostEval Setup Summary

Sprung mass 226.4 kg | CG (-0.810, 0.000, 0.292) m | Wheelbase 1.549 m | Track Fr/Rr 1.212 / 1.212 m
CG bias Fr/Rr 47.7 / 52.3% | CG bias L/R 50.0 / 50.0%

Front Setup				Rear Setup			
FL / FR				RL / RR			
Metric	FL	FR	Units	Metric	RL	RR	Units
Sprung mass	54.0	54.0	kg	Sprung mass	59.2	59.2	kg
Unsprung mass	9.0	9.0	kg	Unsprung mass	8.3	8.3	kg
Sprung load	530	530	N	Sprung load	580	580	N
Motion ratio	1.005	1.005	wheel/spring	Motion ratio	1.239	1.239	wheel/spring
Spring rate	26269	26269	N/m	Spring rate	43782	43782	N/m
Spring force	532	532	N	Spring force	719	719	N
Installed length	179.4	179.4	mm	Installed length	247.0	247.0	mm
Compression	20.3	20.3	mm	Compression	16.4	16.4	mm
Free length	199.7	199.7	mm	Free length	263.4	263.4	mm
Balance target	529.7	529.7	N	Balance target	580.5	580.5	N
Static Fz	529.8	529.6	N	Static Fz	580.6	580.4	N
Fz error	+0.1	-0.1	N	Fz error	+0.1	-0.1	N
Wheel rate	25992	25992	N/m	Wheel rate	28535	28535	N/m
Sprung freq	3.10	3.10	Hz	Sprung freq	3.07	3.07	Hz
Unsprung freq	18.79	18.79	Hz	Unsprung freq	19.80	19.80	Hz

FourPostEval Metrics Summary

Heave Gains

Camber	0.0261	deg/mm
Toe	-0.0001	deg/mm
Caster	-0.0022	deg/mm
KPI	0.0261	deg/mm
Trail	-0.0011	mm/mm
Scrub	0.0018	mm/mm

Roll Gains

Camber	0.7246	deg/deg
Toe	0.0014	deg/deg
Caster	0.0344	deg/deg
KPI	0.7250	deg/deg
Trail	-0.0261	mm/deg
Scrub	0.0454	mm/deg

Anti Metrics

Anti-Dive	2.8	%
Anti-Squat	-4.6	%
Front Anti-Roll	0.1	%
Rear Anti-Roll	0.1	%
LLTD (Front)	49.1	%

Motion Ratios

Front MR	0.987	wheel/spring
Rear MR	1.253	wheel/spring
Front Bar MR	0.128	roll/stabar
Rear Bar MR	0.155	roll/stabar

Setup Check

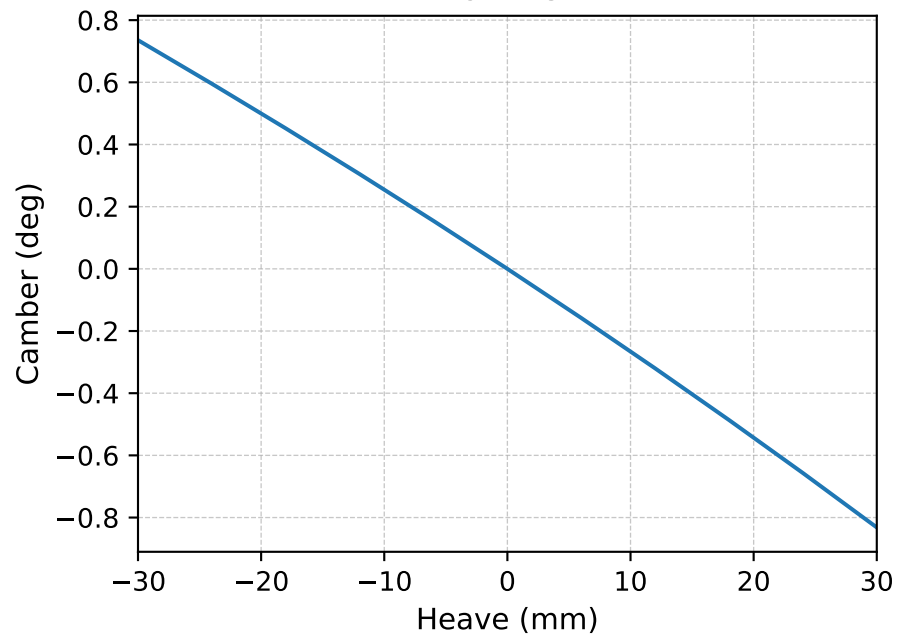
Max Static Fz Error	0.0	N
Max Static Fz Error	0.00	%

Roll Stiffness

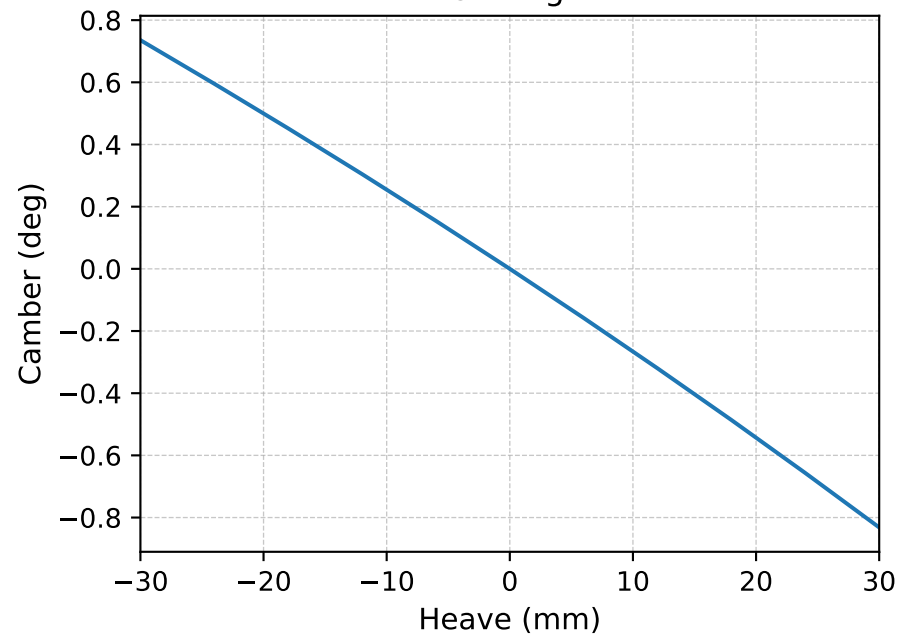
Spring Front	346	Nm/deg
Spring Rear	358	Nm/deg
ARB Front	412	Nm/deg
ARB Rear	383	Nm/deg
Total Front	758	Nm/deg
Total Rear	740	Nm/deg

Camber vs Heave

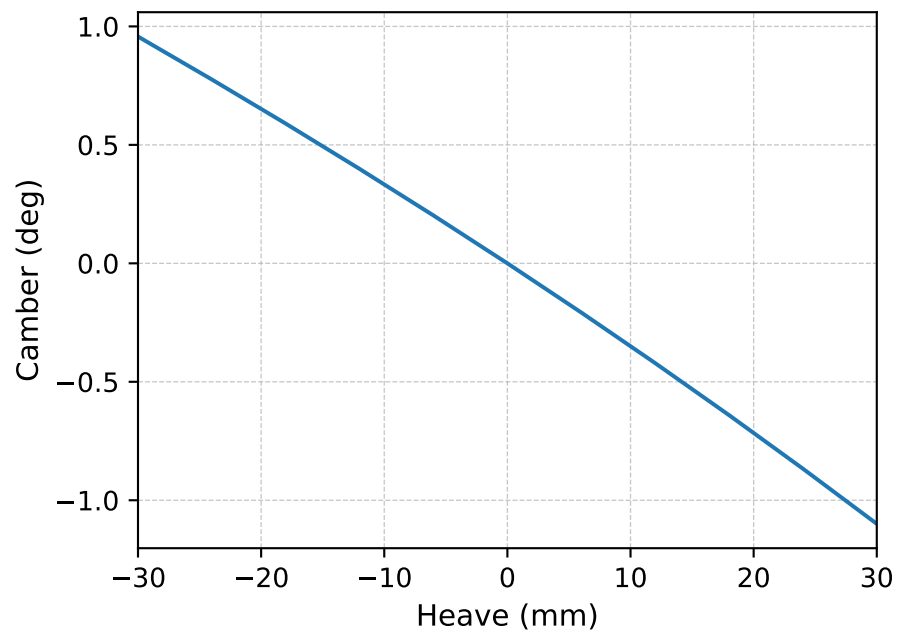
Front Left



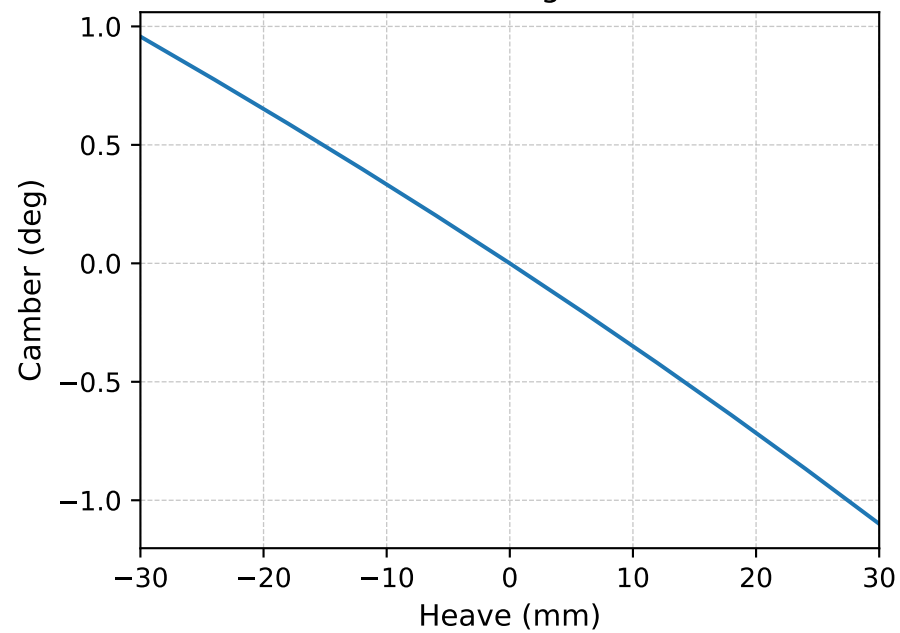
Front Right



Rear Left

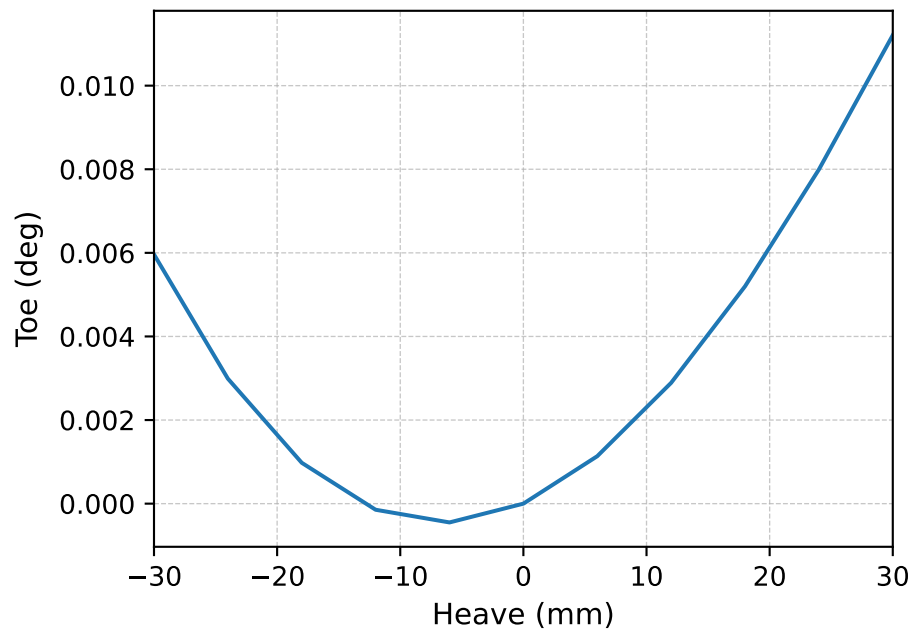


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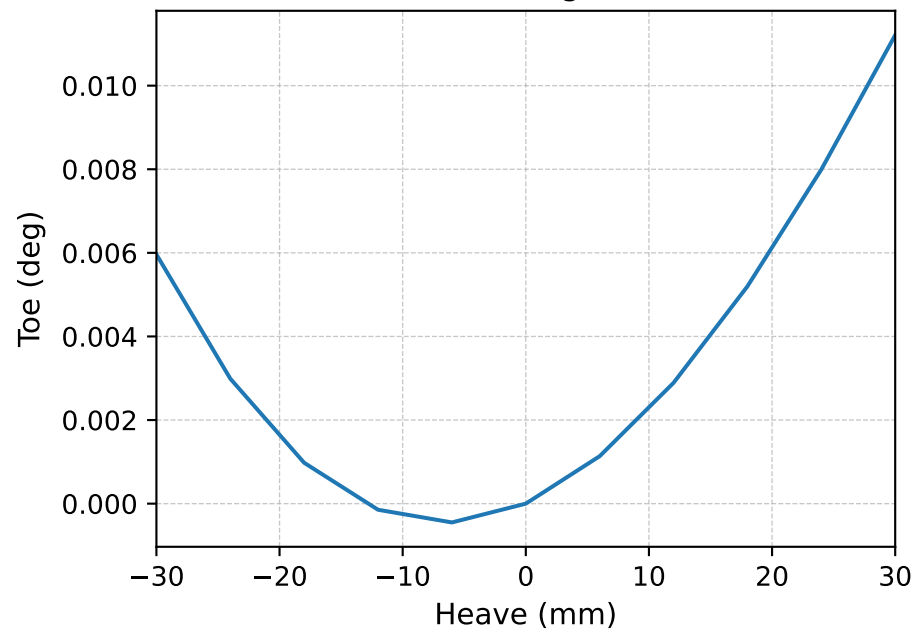


Toe vs Heave

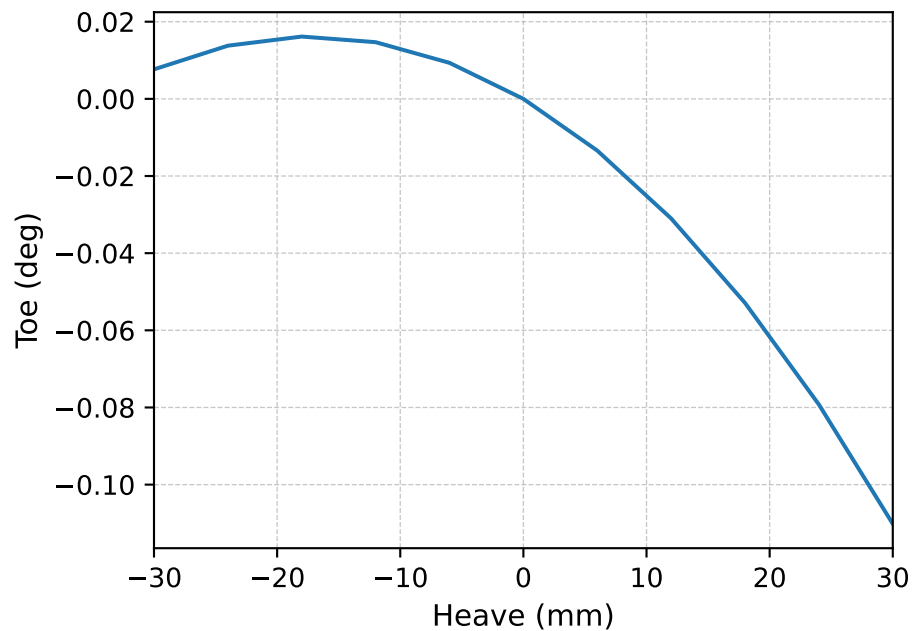
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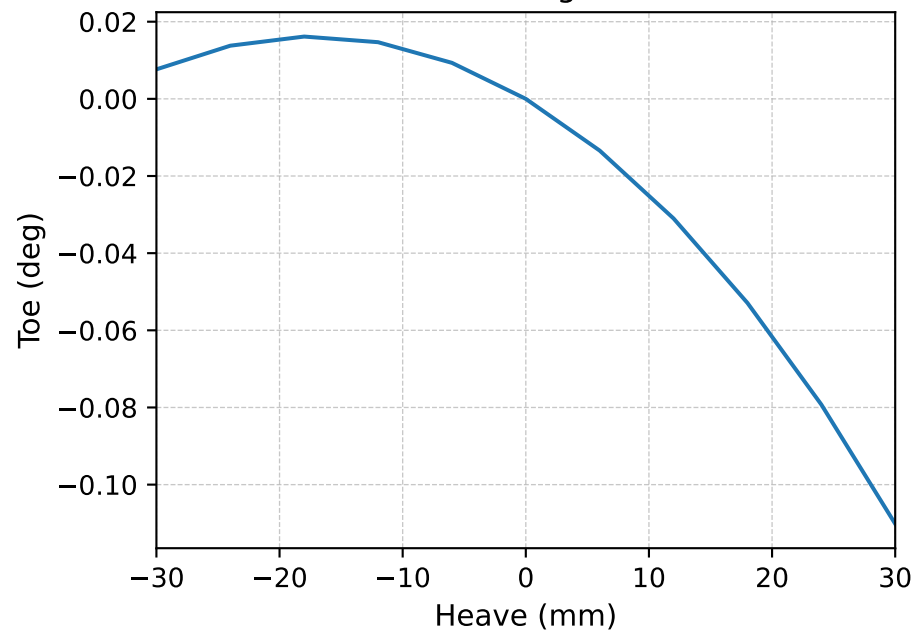
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Rear Left

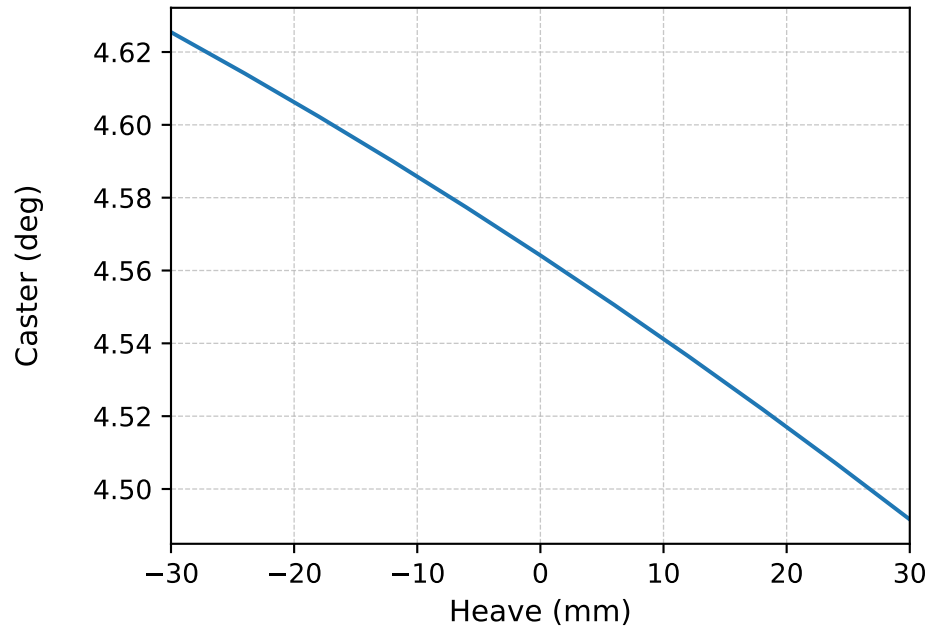


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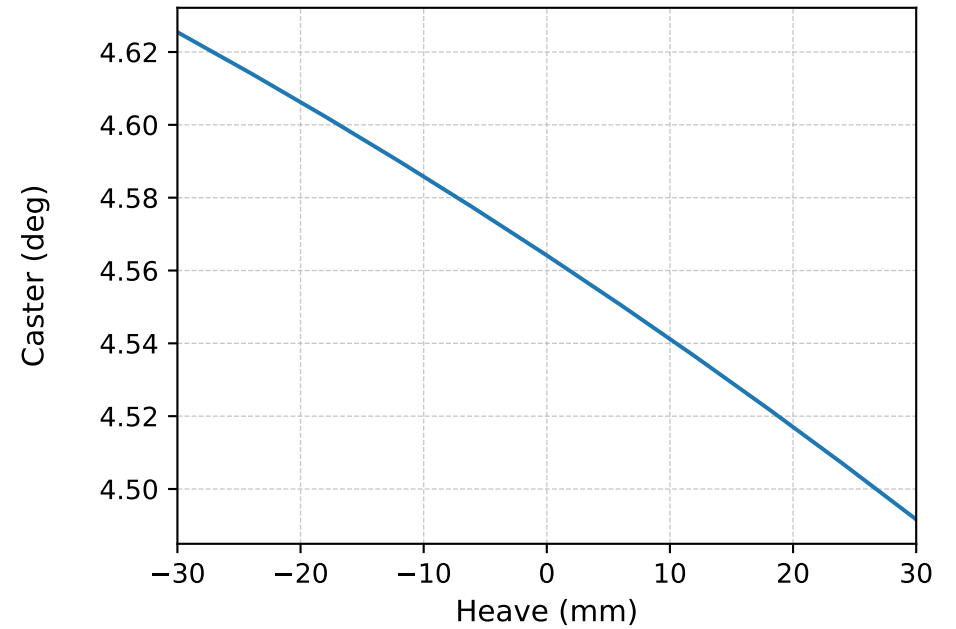


Caster vs Heave

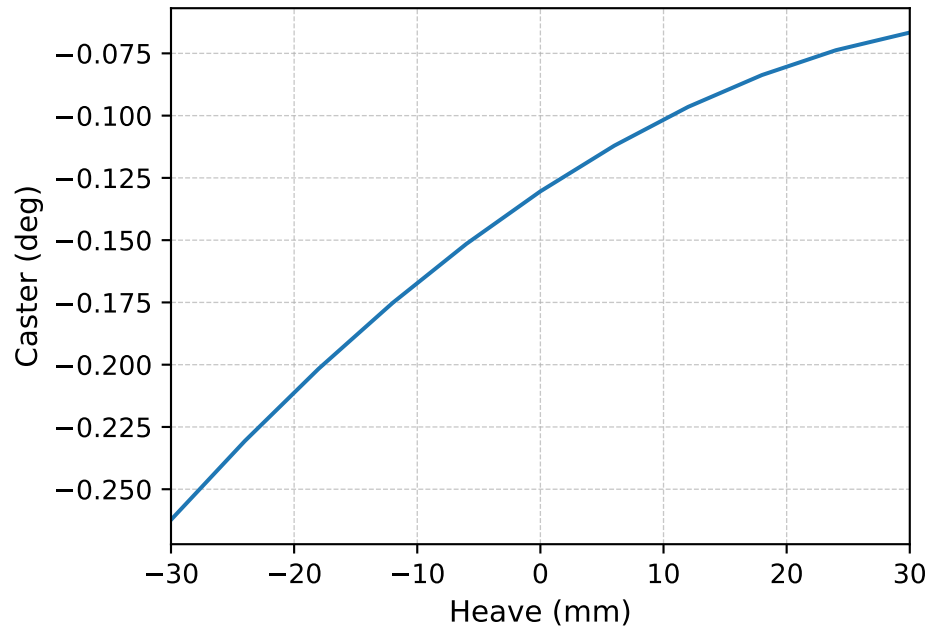
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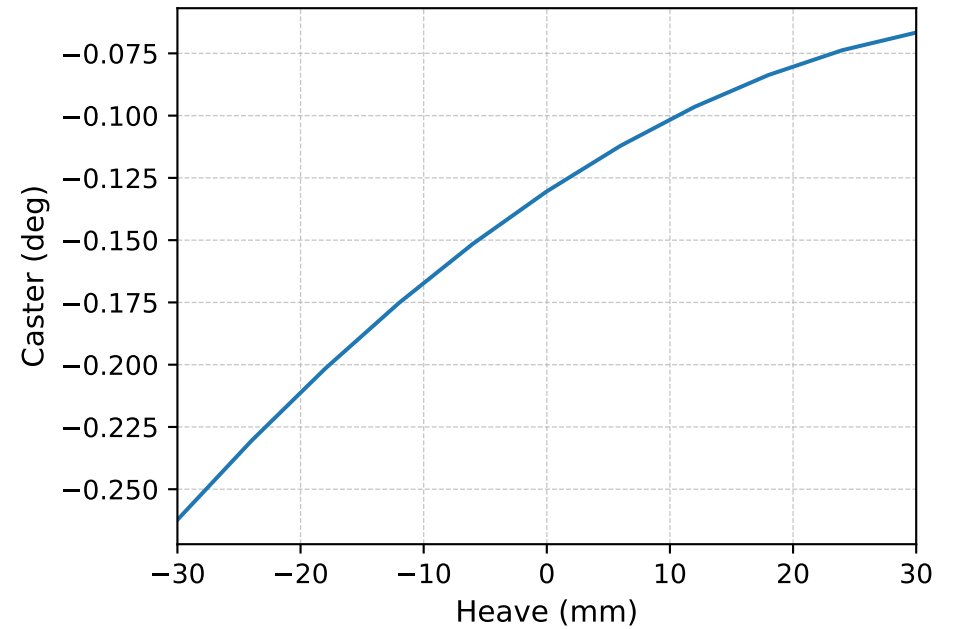
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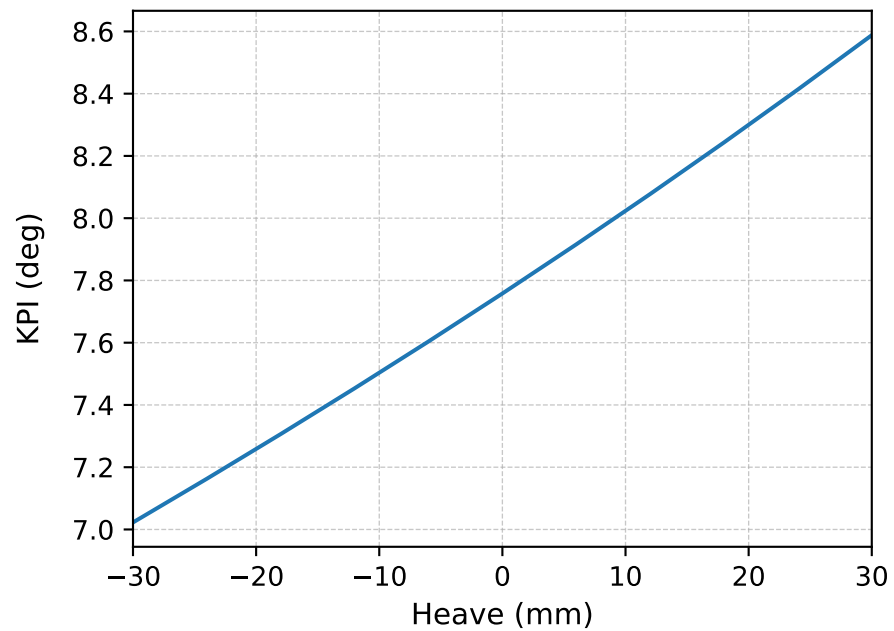


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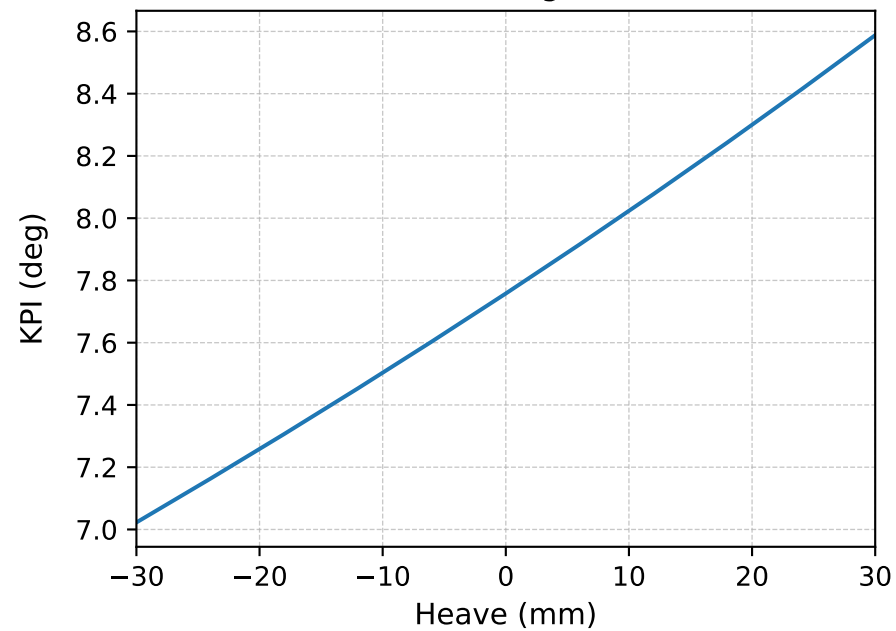


KPI vs Heave

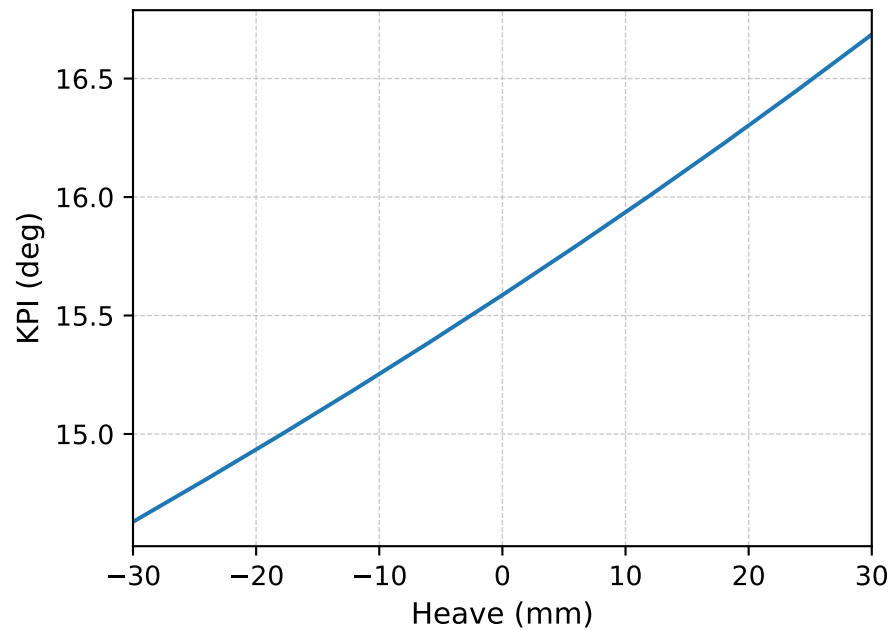
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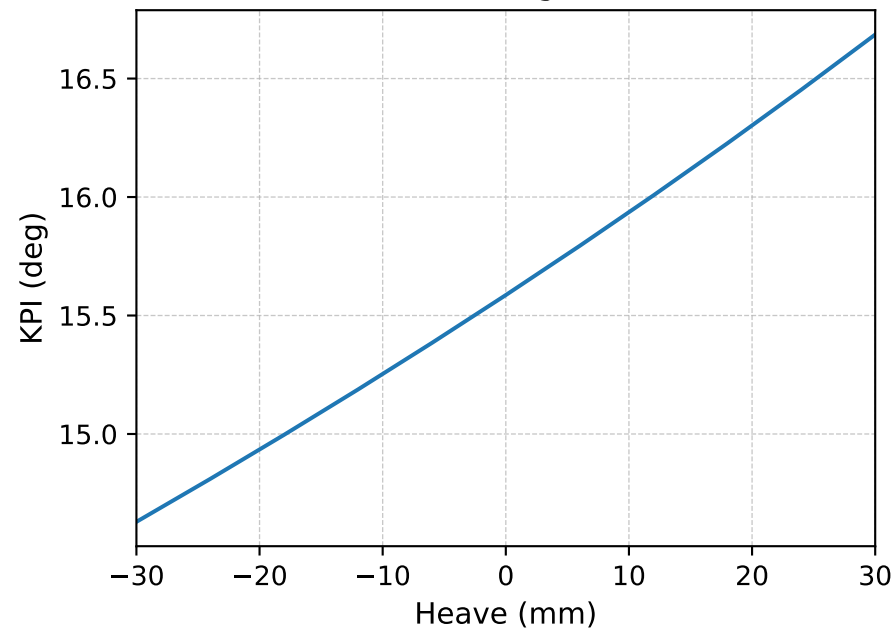
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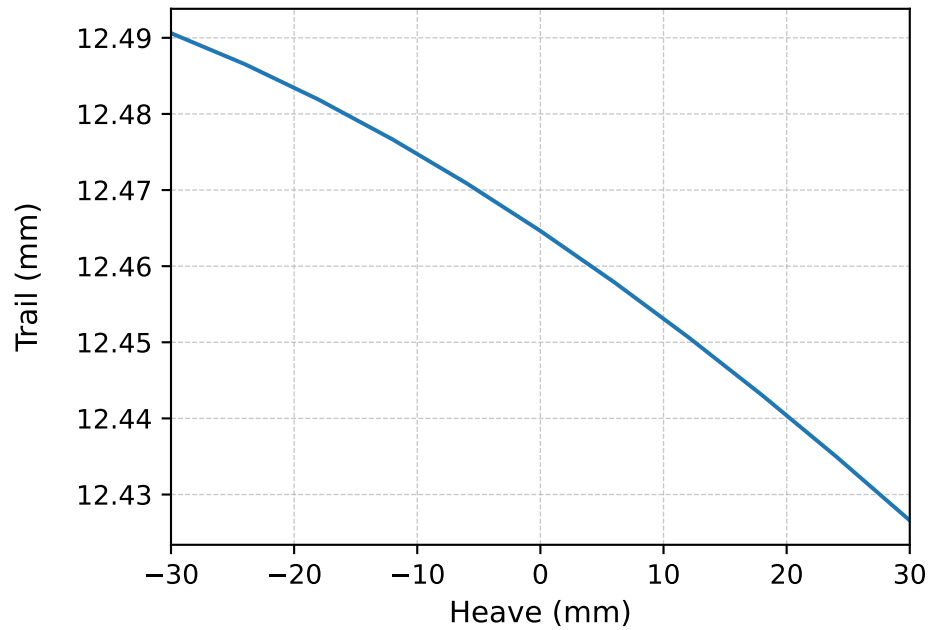


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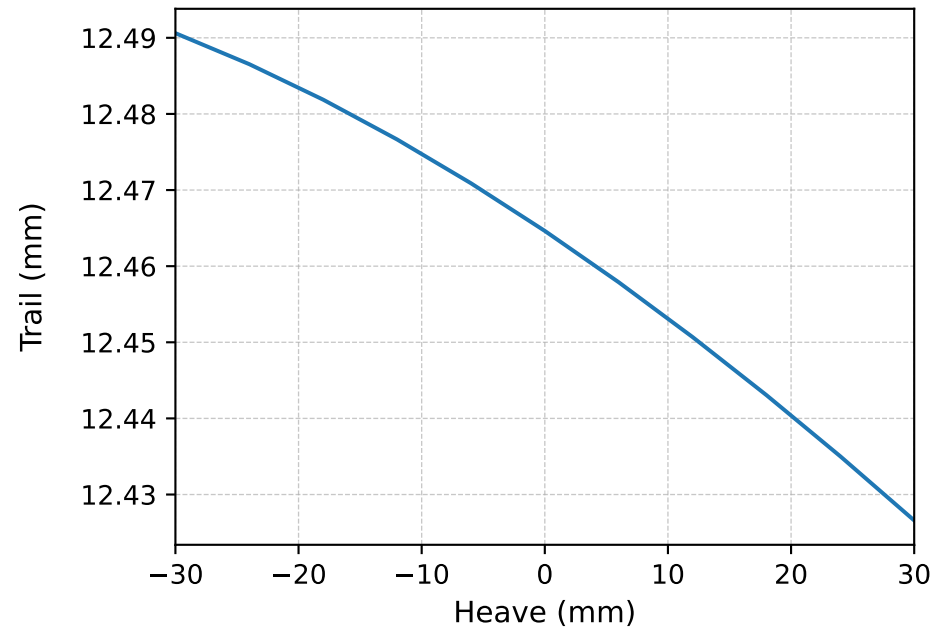


Mechanical Trail vs Heave

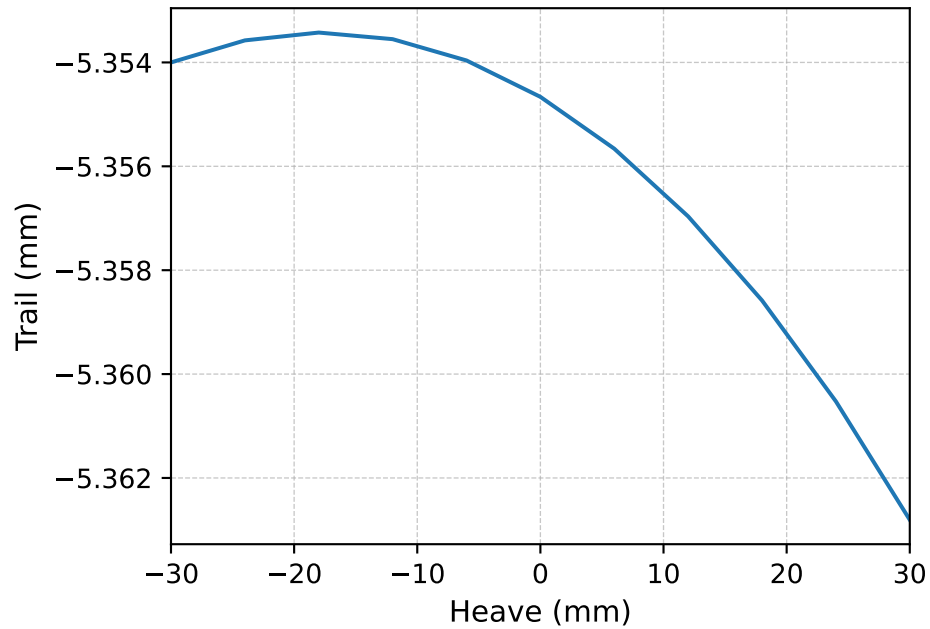
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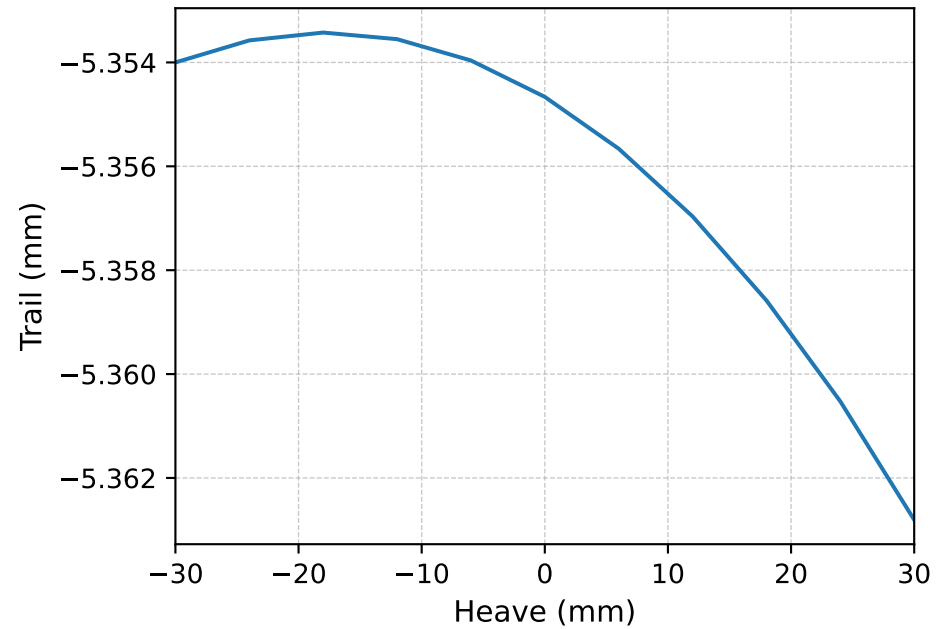
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Rear Left

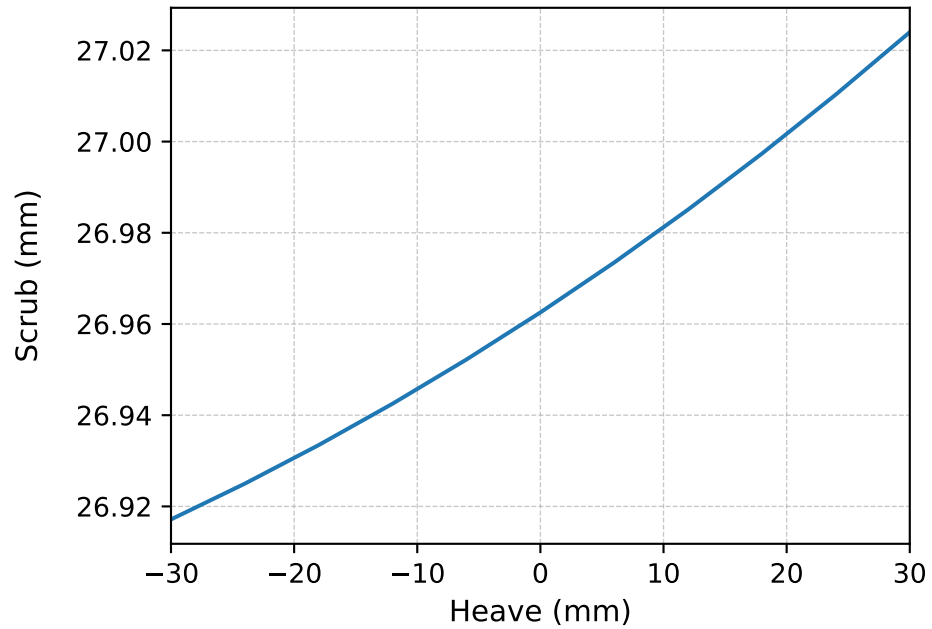


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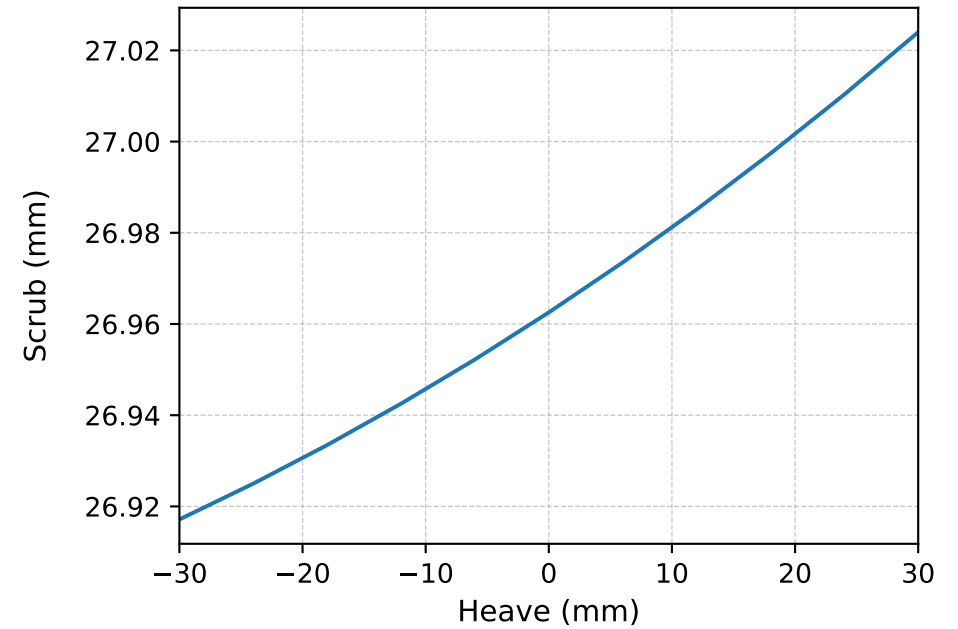


Mechanical Scrub vs Heave

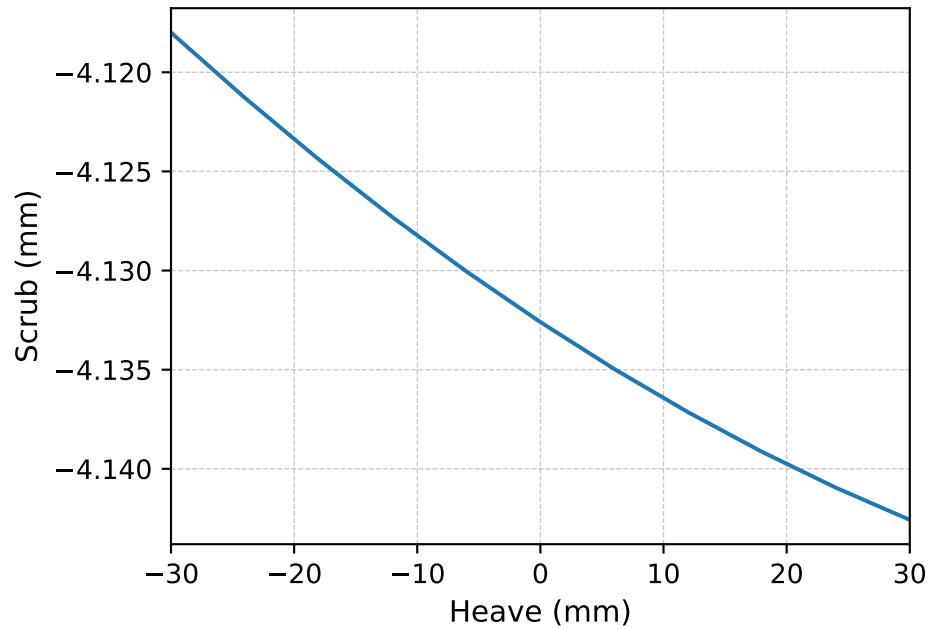
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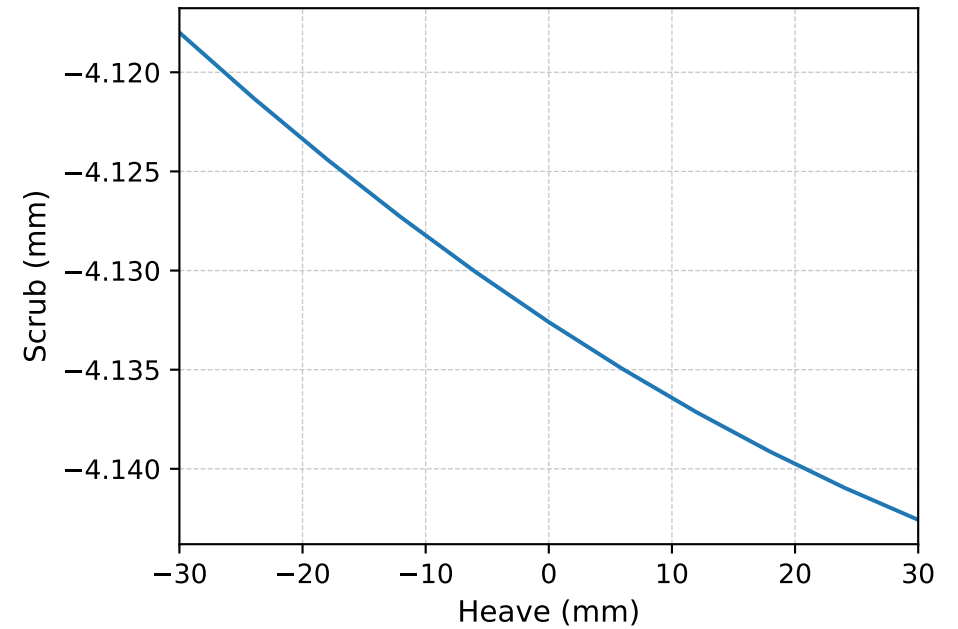
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Rear Left

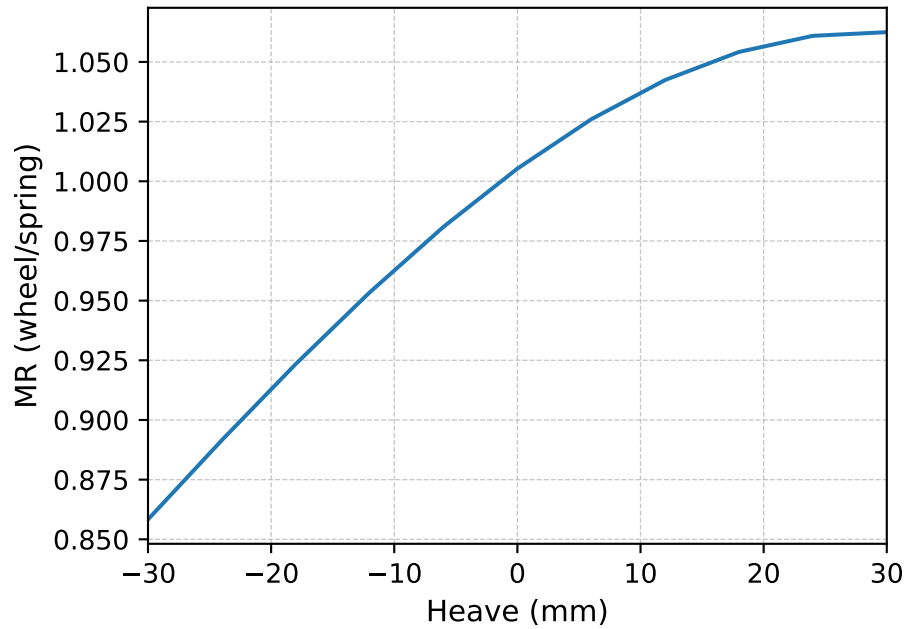


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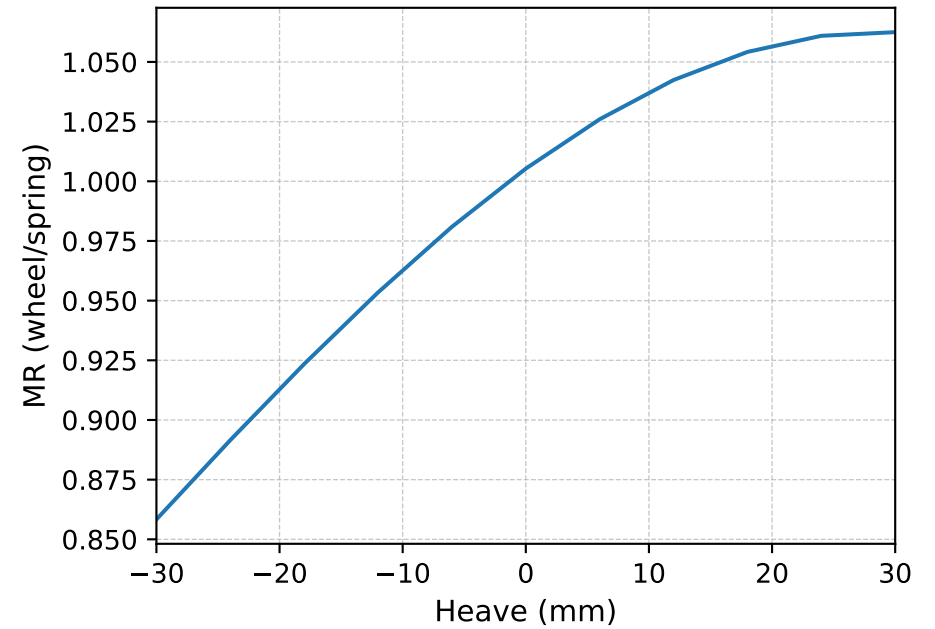


Motion Ratio vs Heave

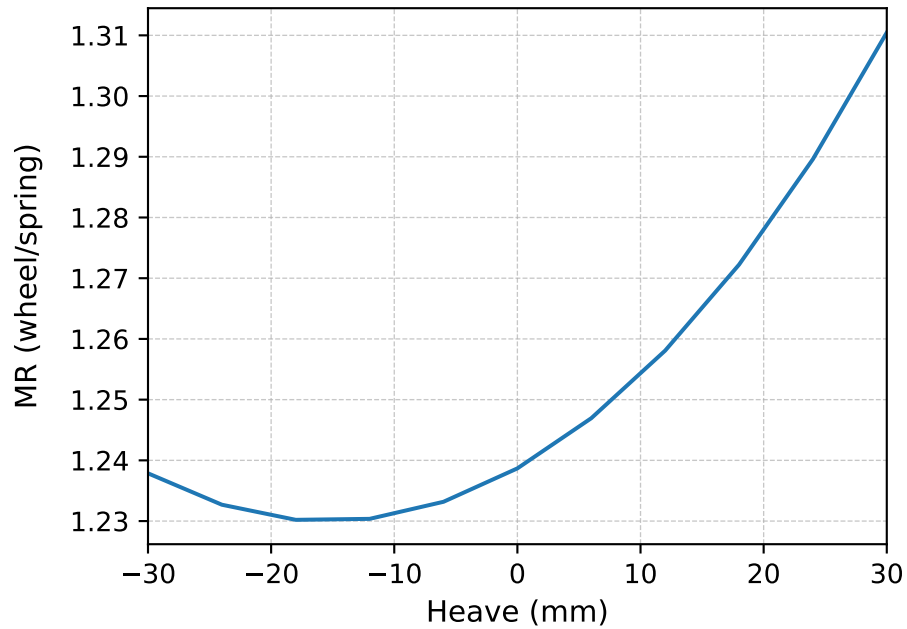
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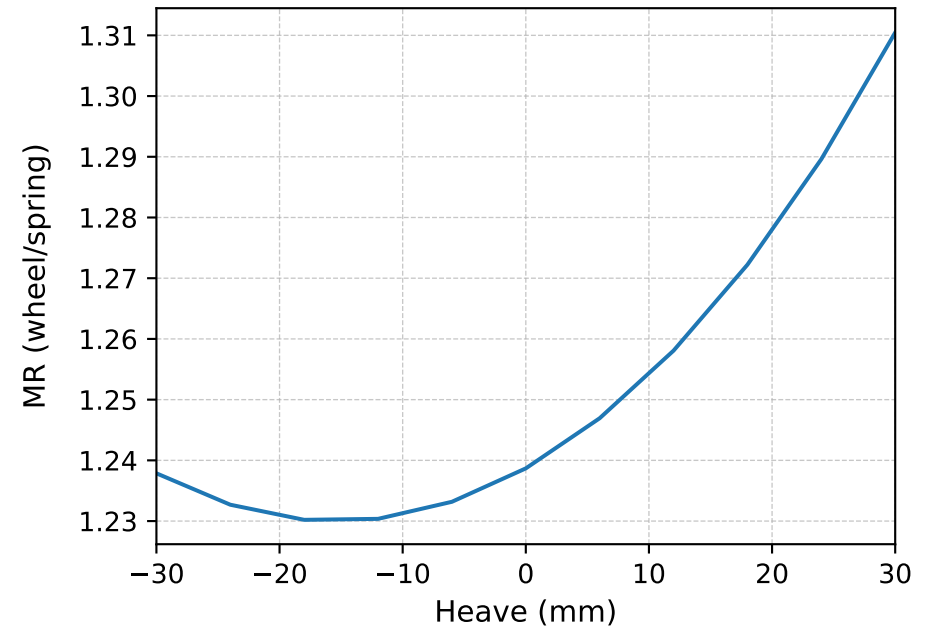
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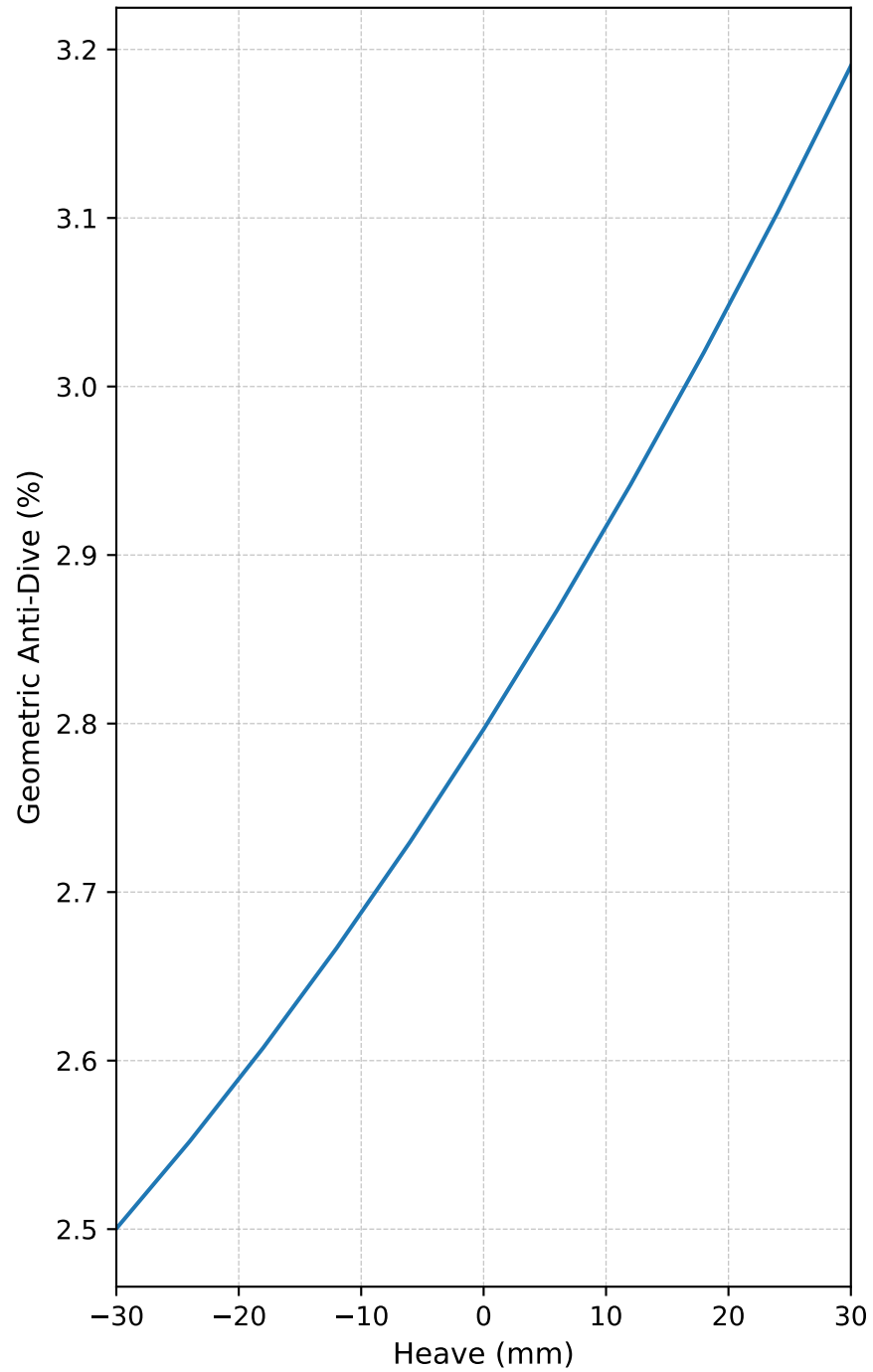


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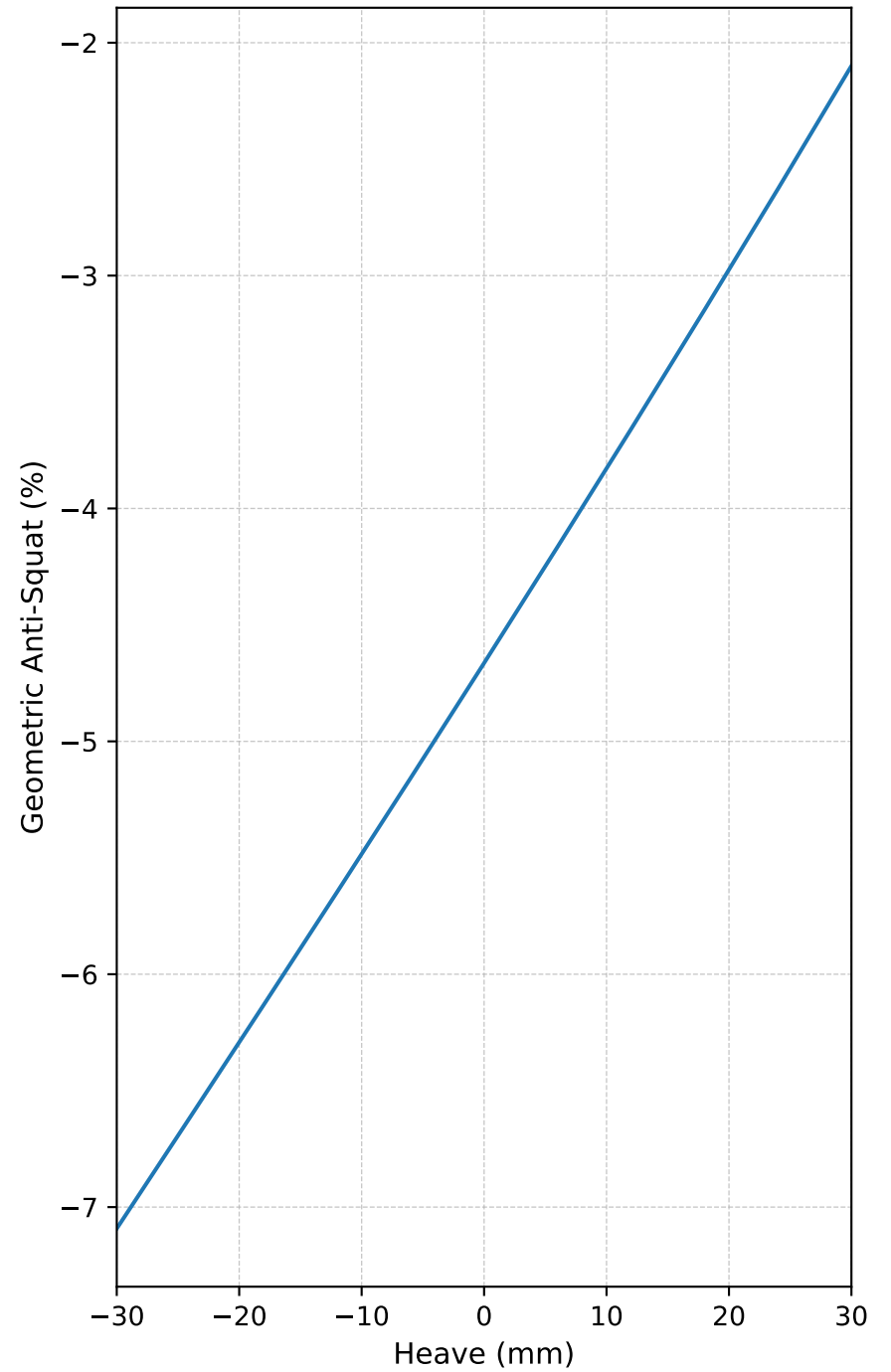


Anti vs Heave

Front

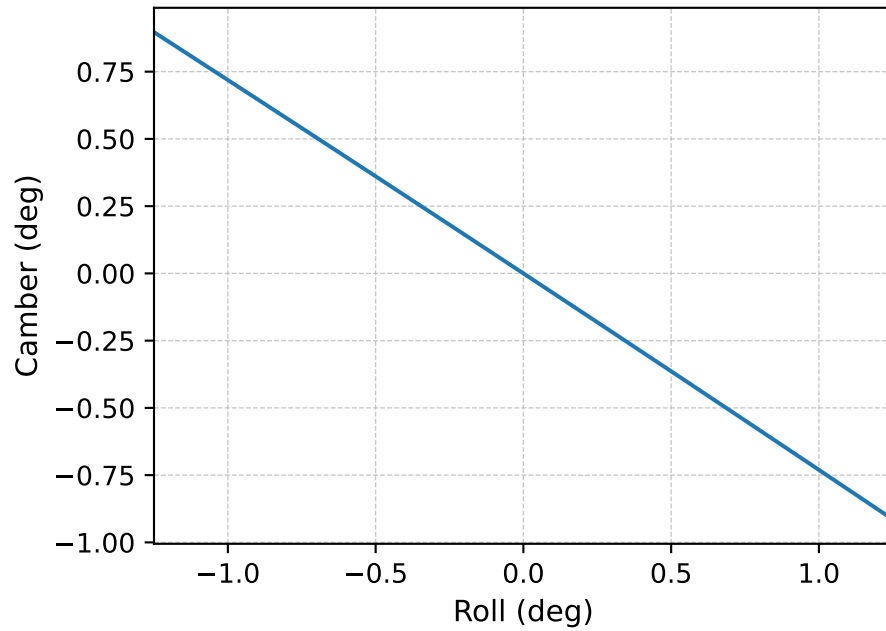


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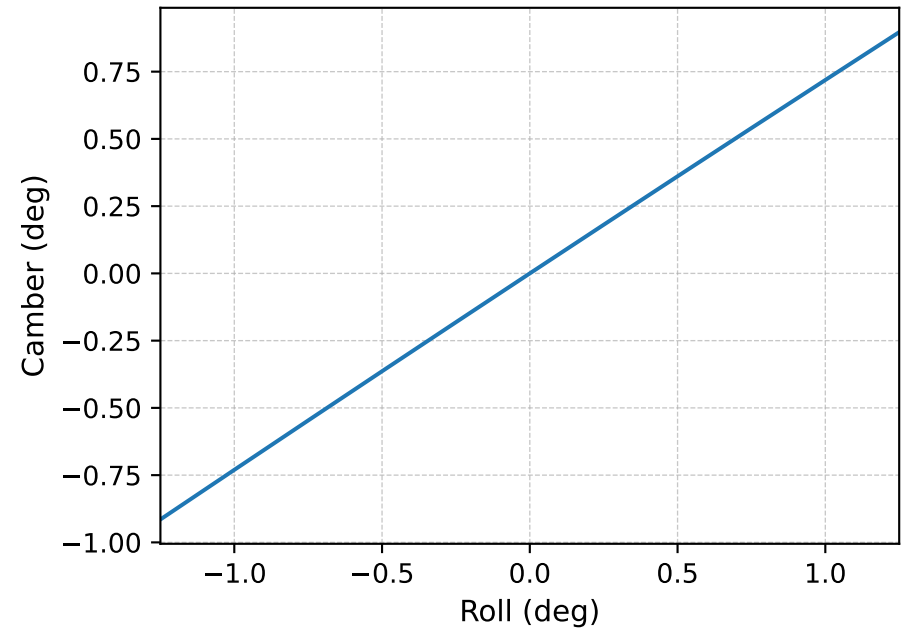


Camber vs Roll

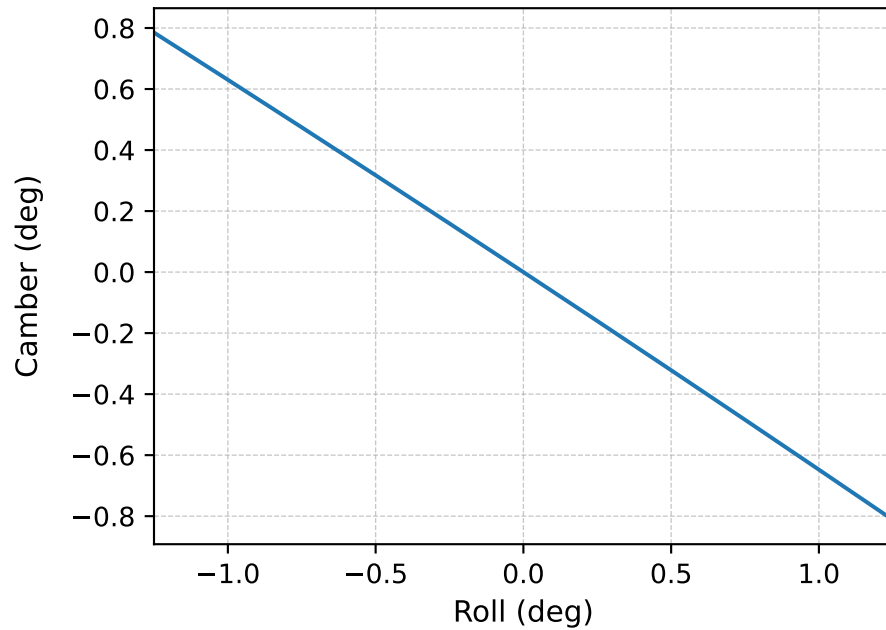
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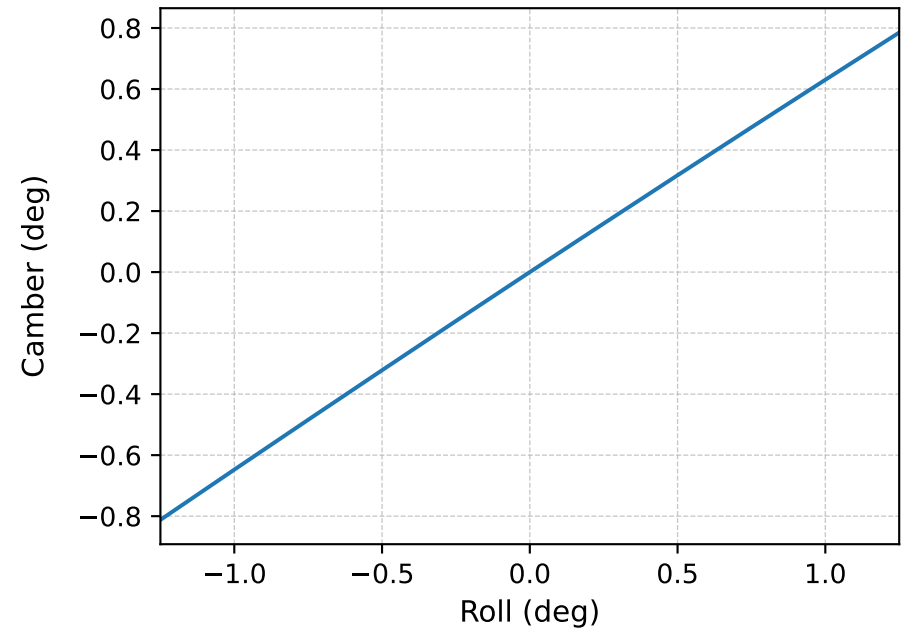
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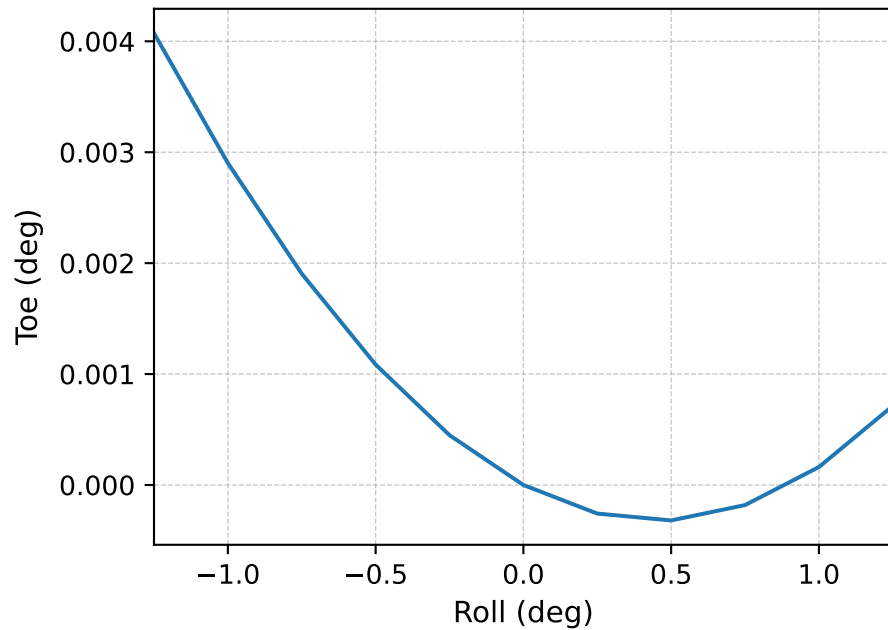


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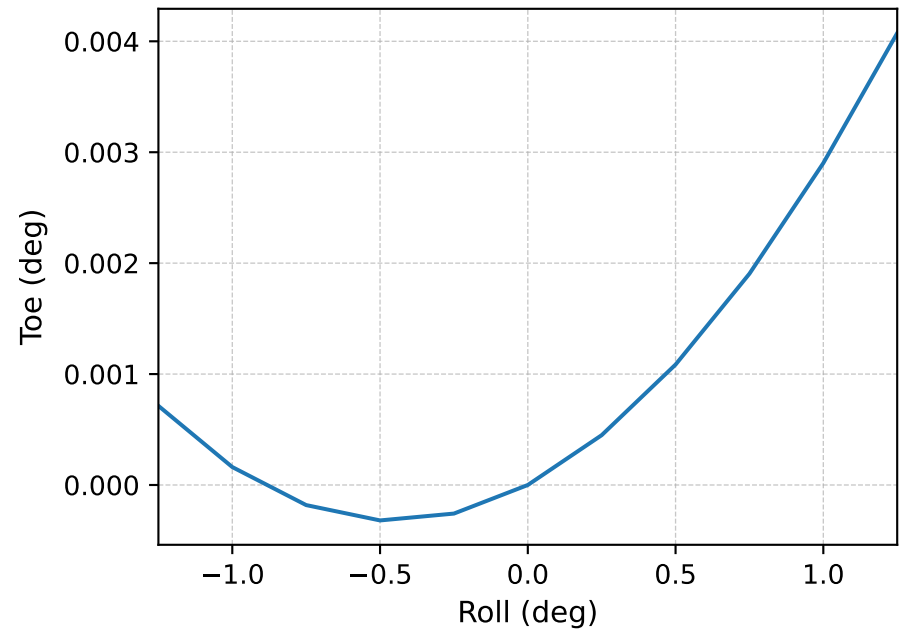


Toe vs Roll

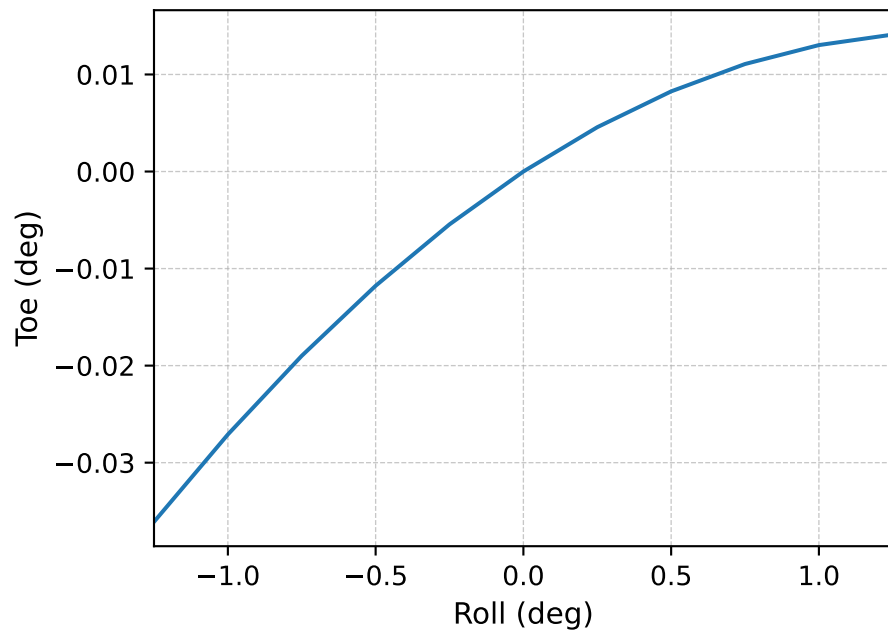
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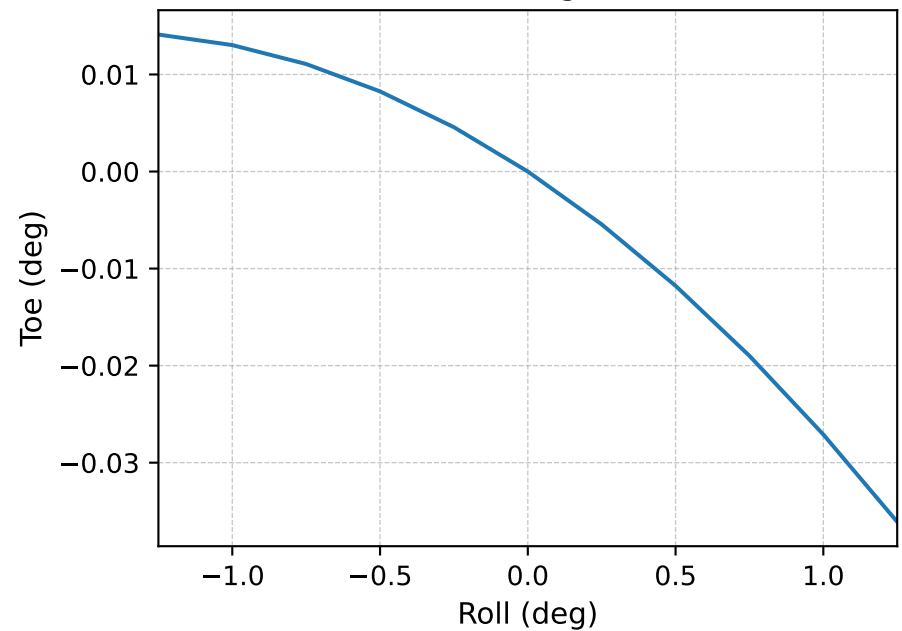
Front Right



Rear Left

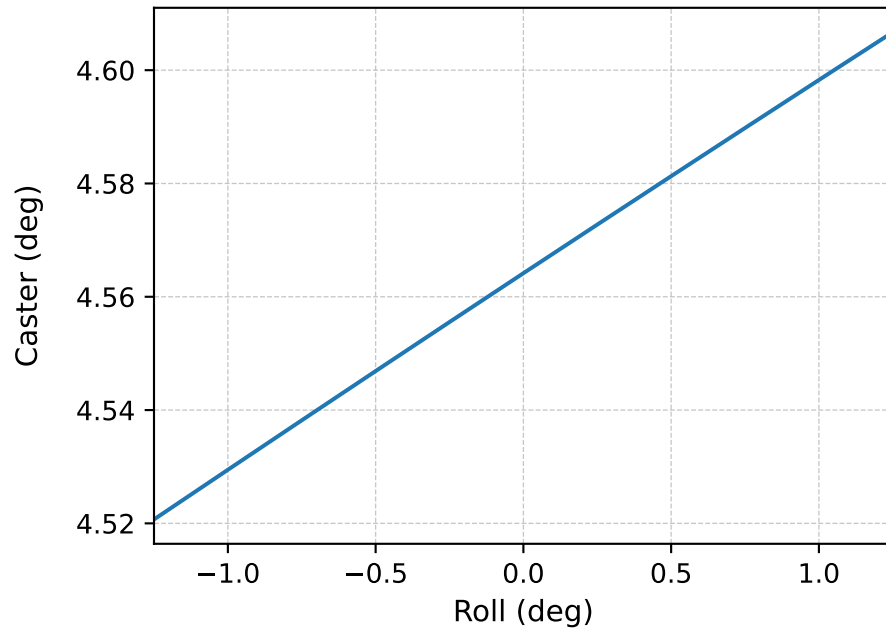


Rear Right

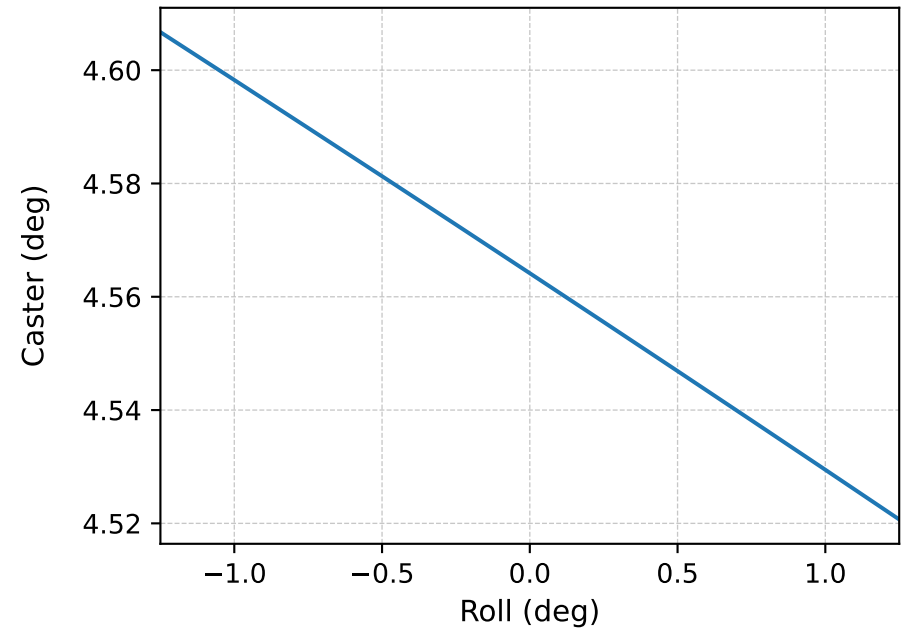


Caster vs Roll

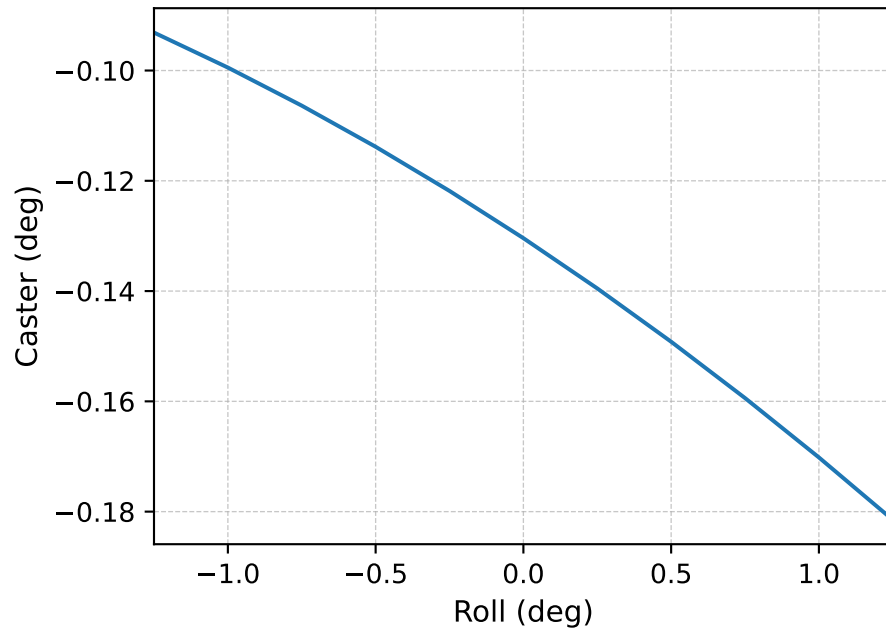
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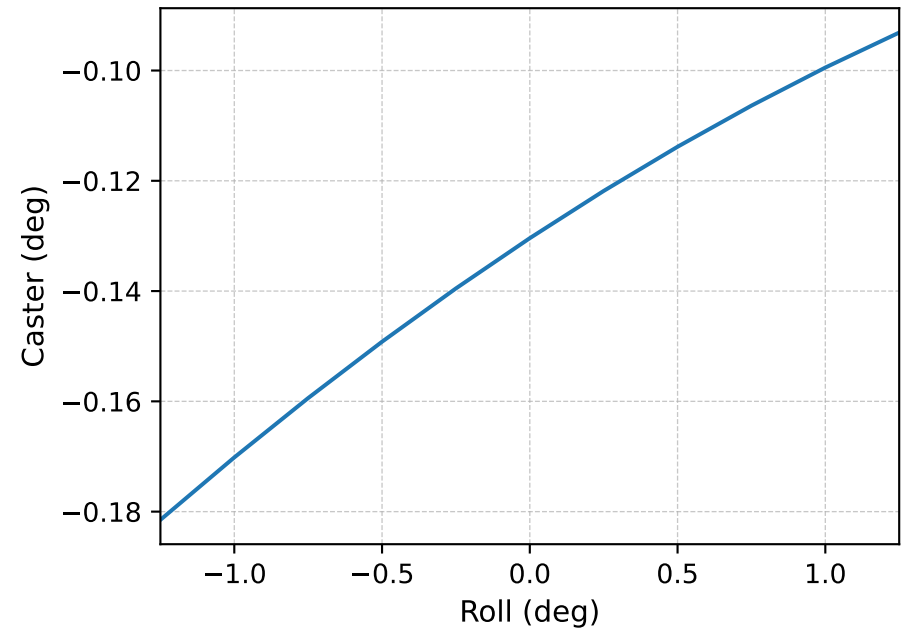
Front Right



Rear Left

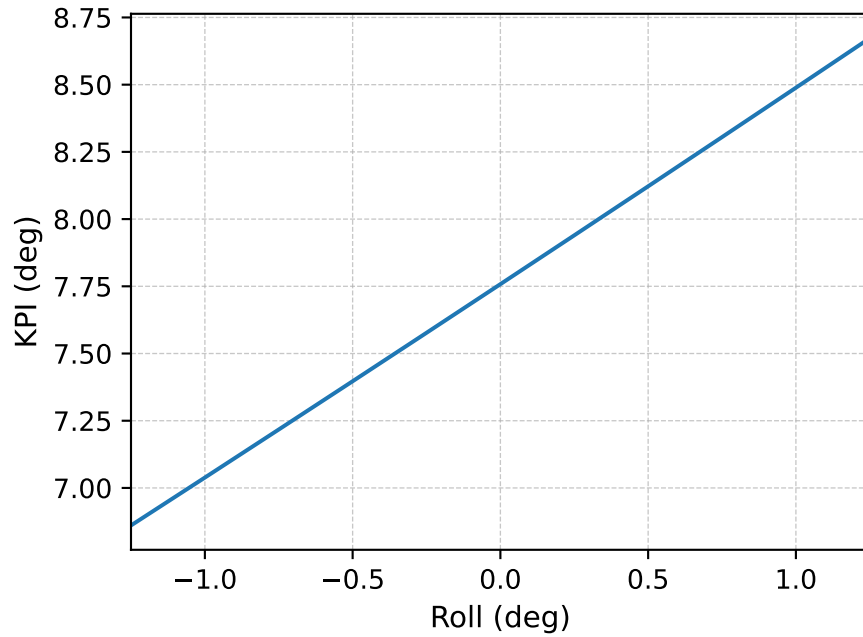


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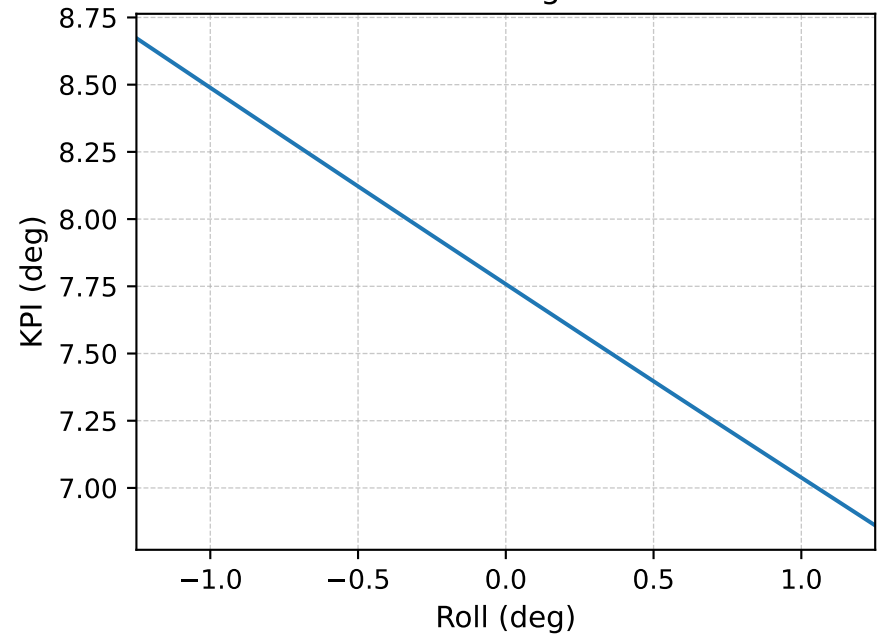


KPI vs Roll

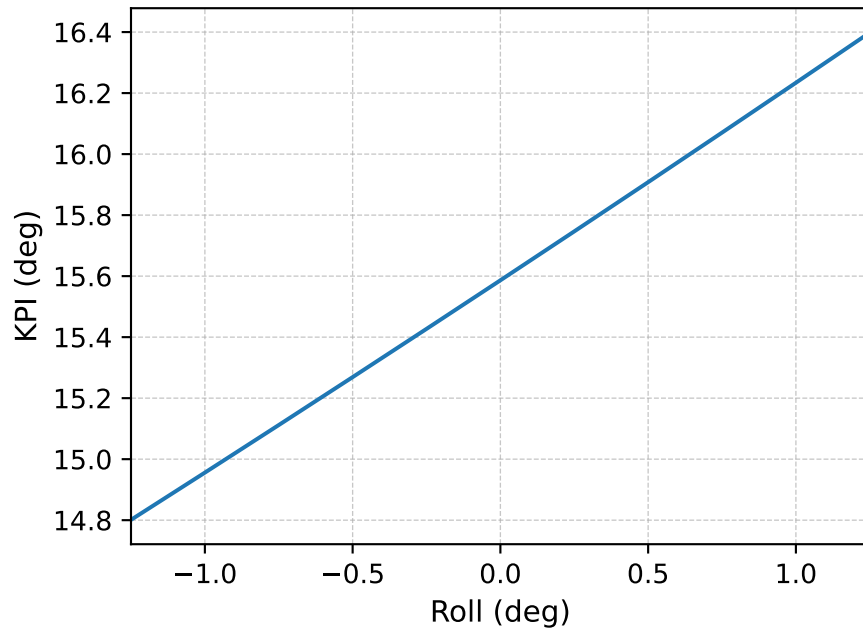
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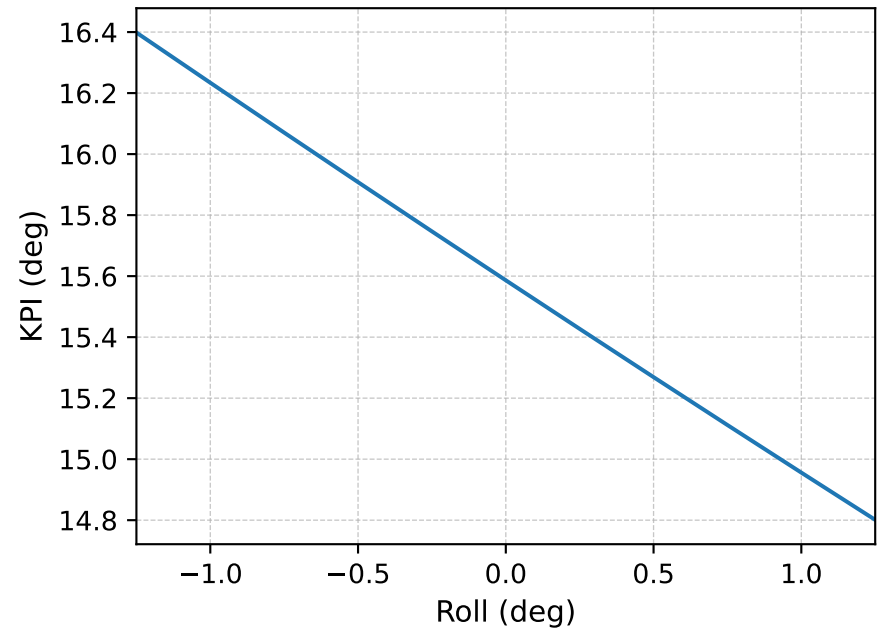
Front Right



Rear Left

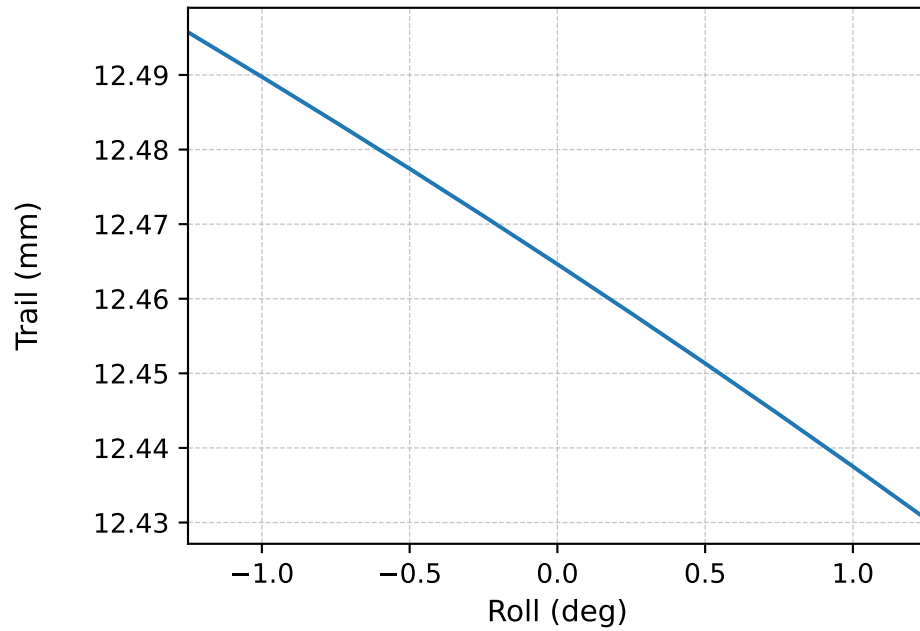


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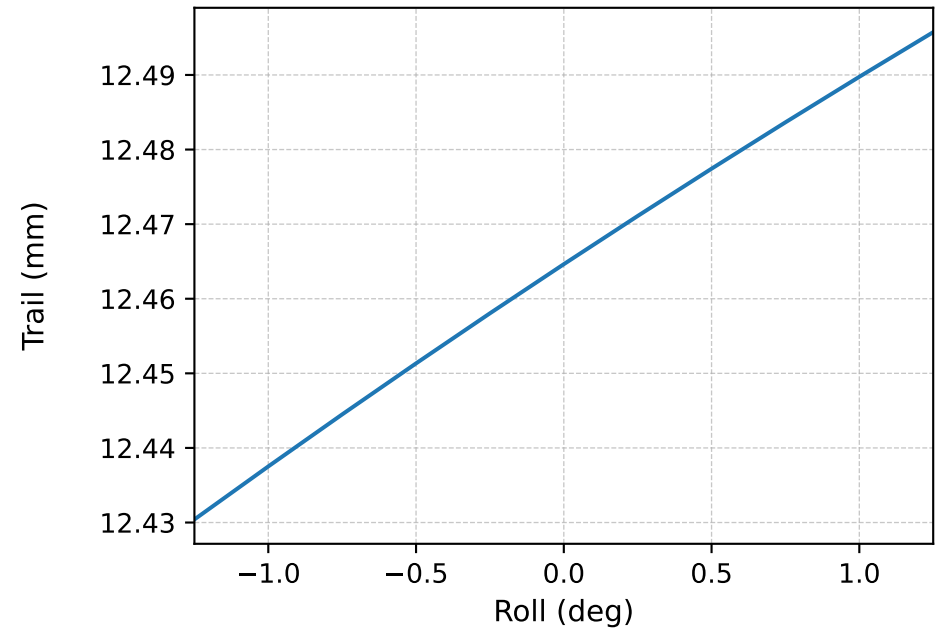


Mechanical Trail vs Roll

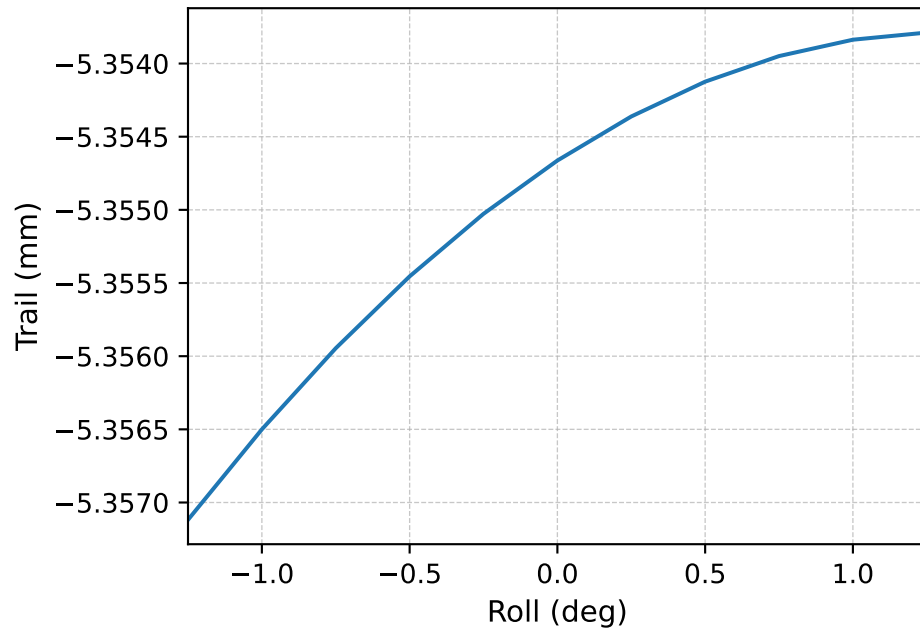
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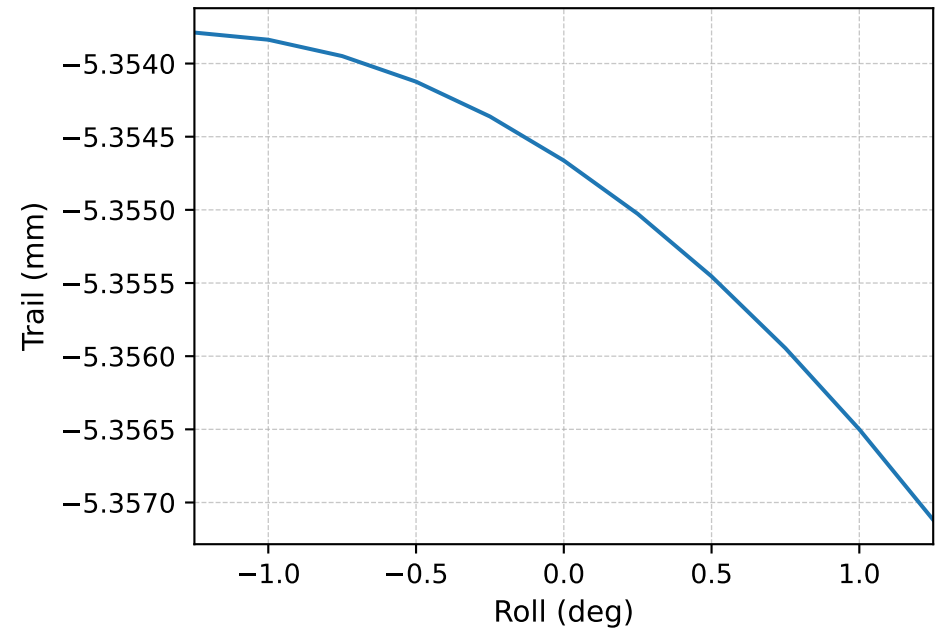
Front Right



Rear Left

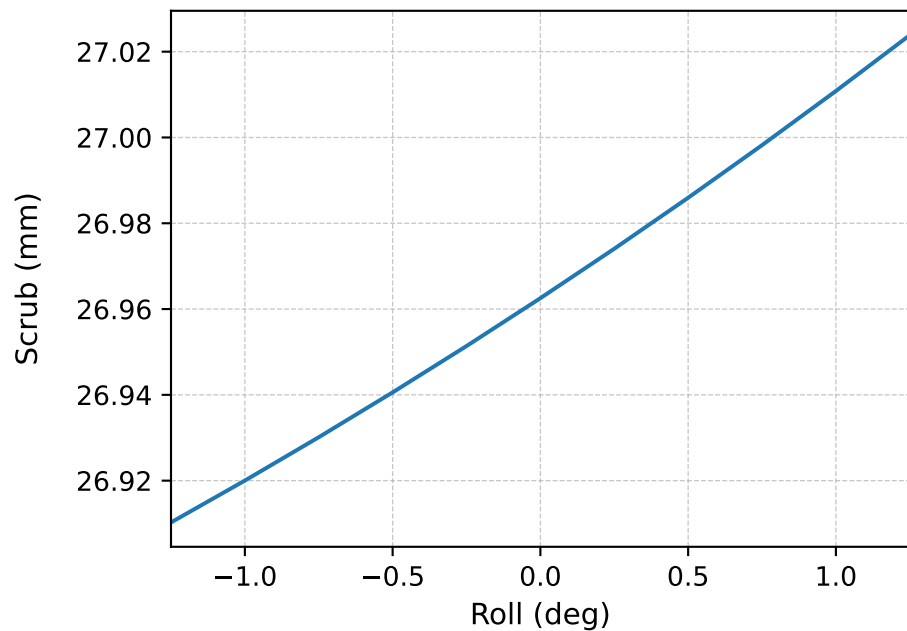


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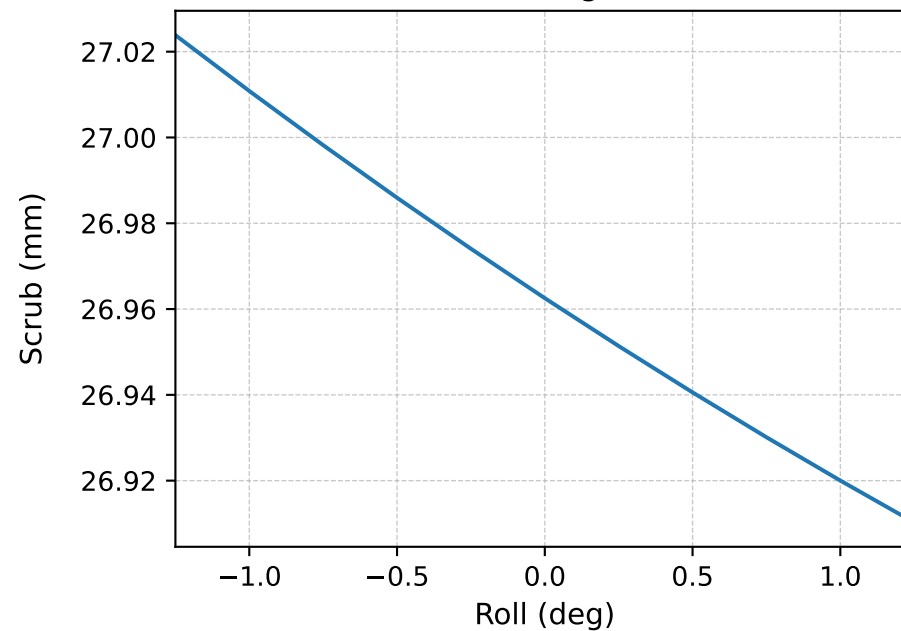


Mechanical Scrub vs Roll

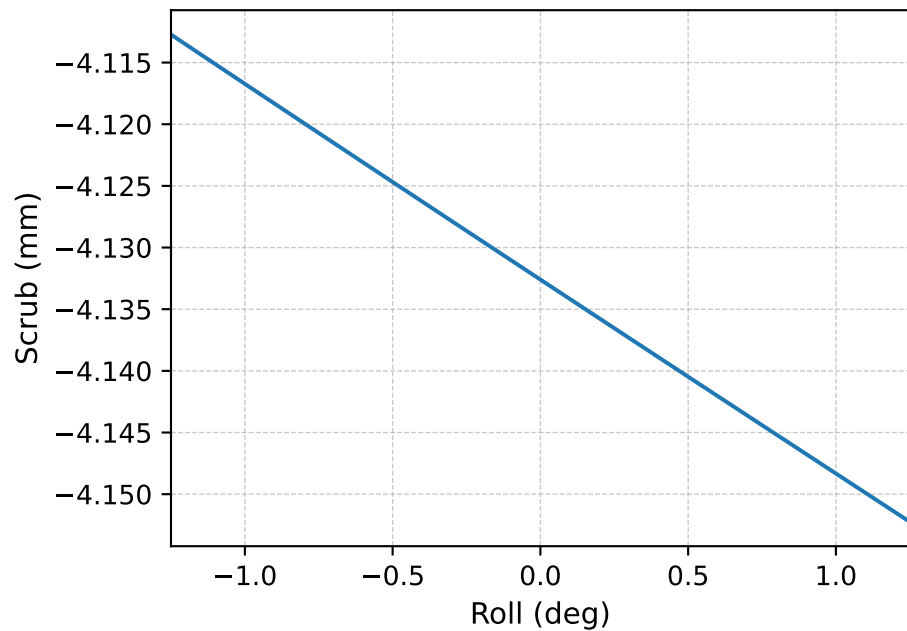
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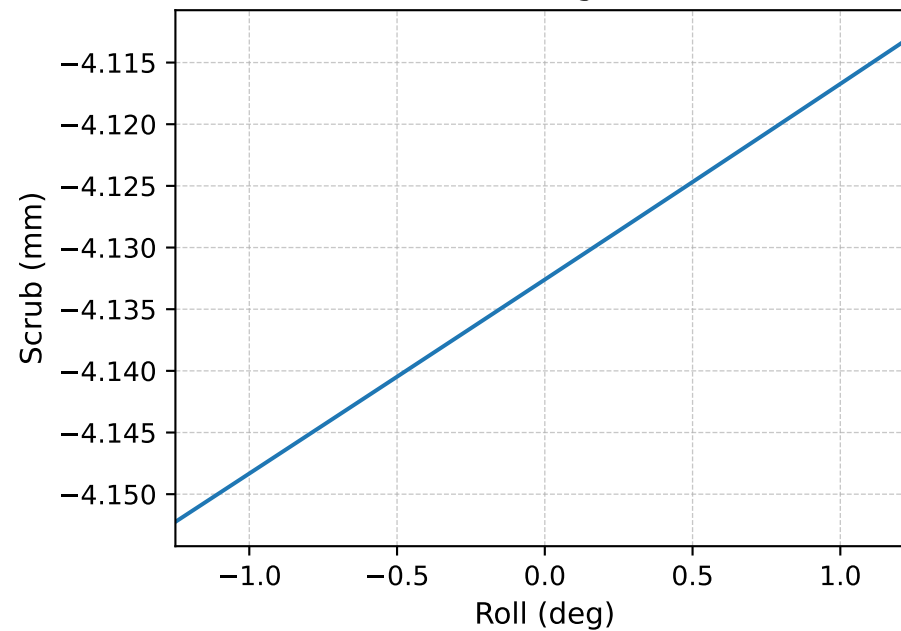
Front Right



Rear Left

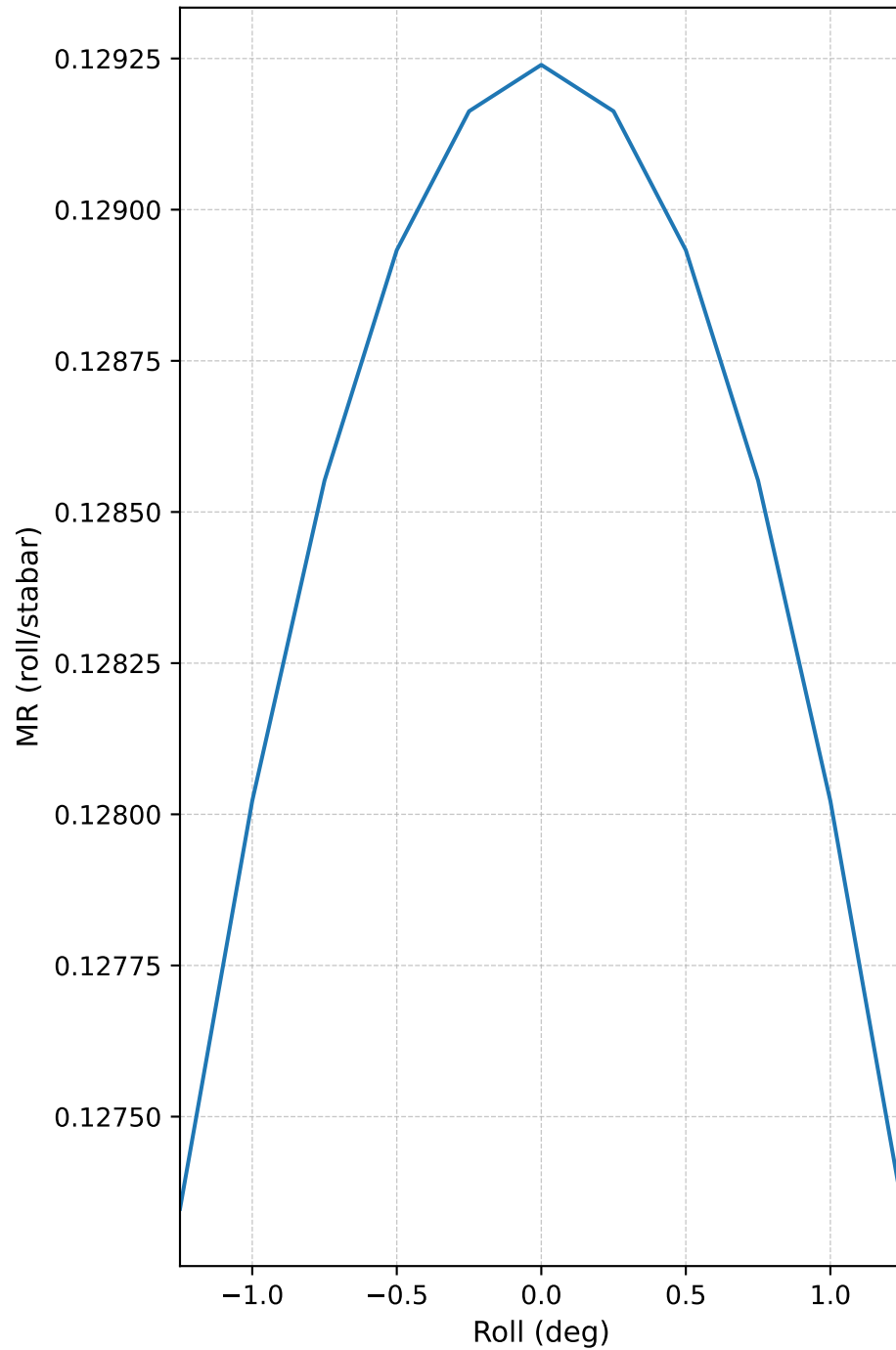


Rear Right

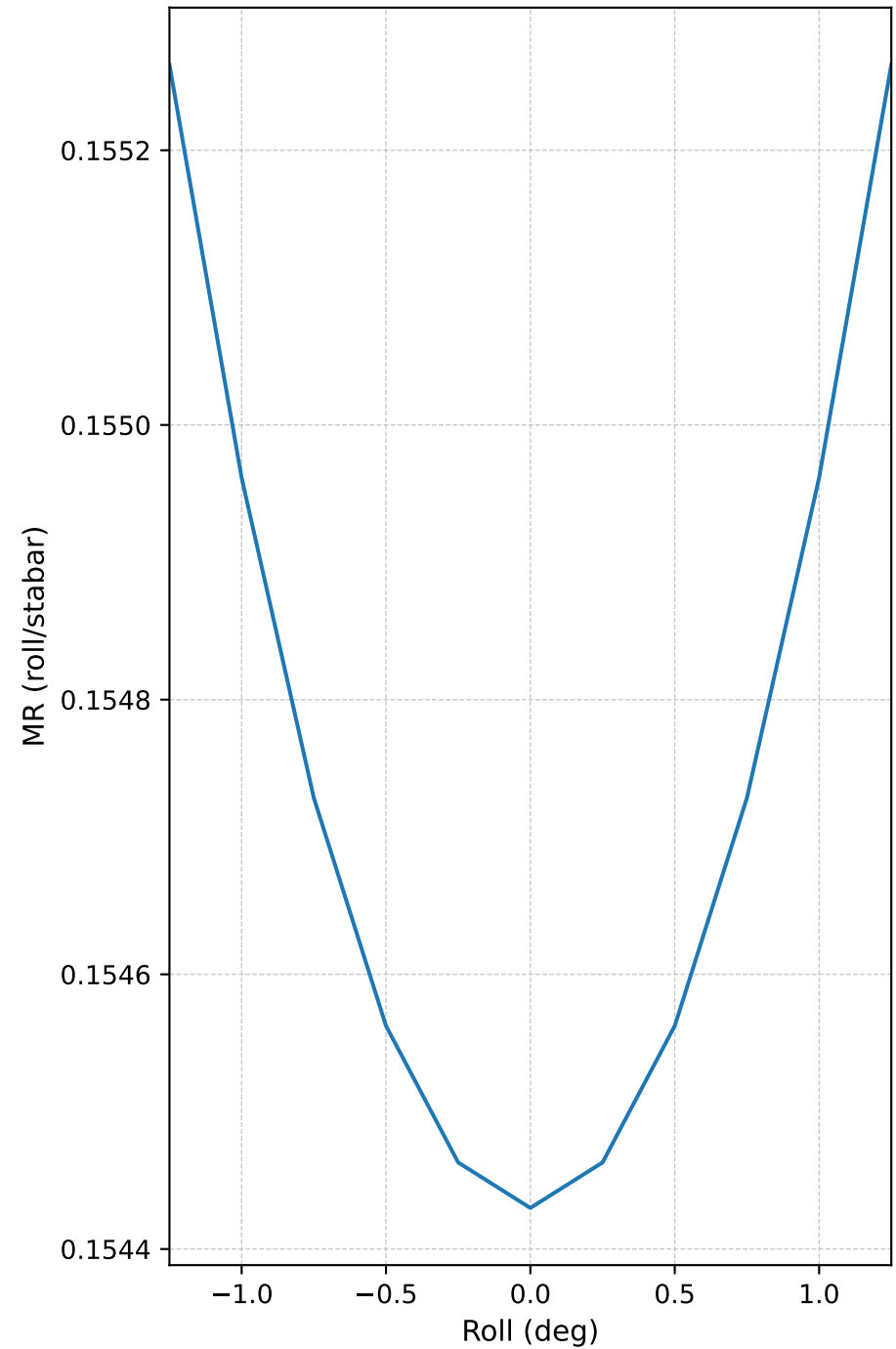


Stabar Motion Ratio vs Roll

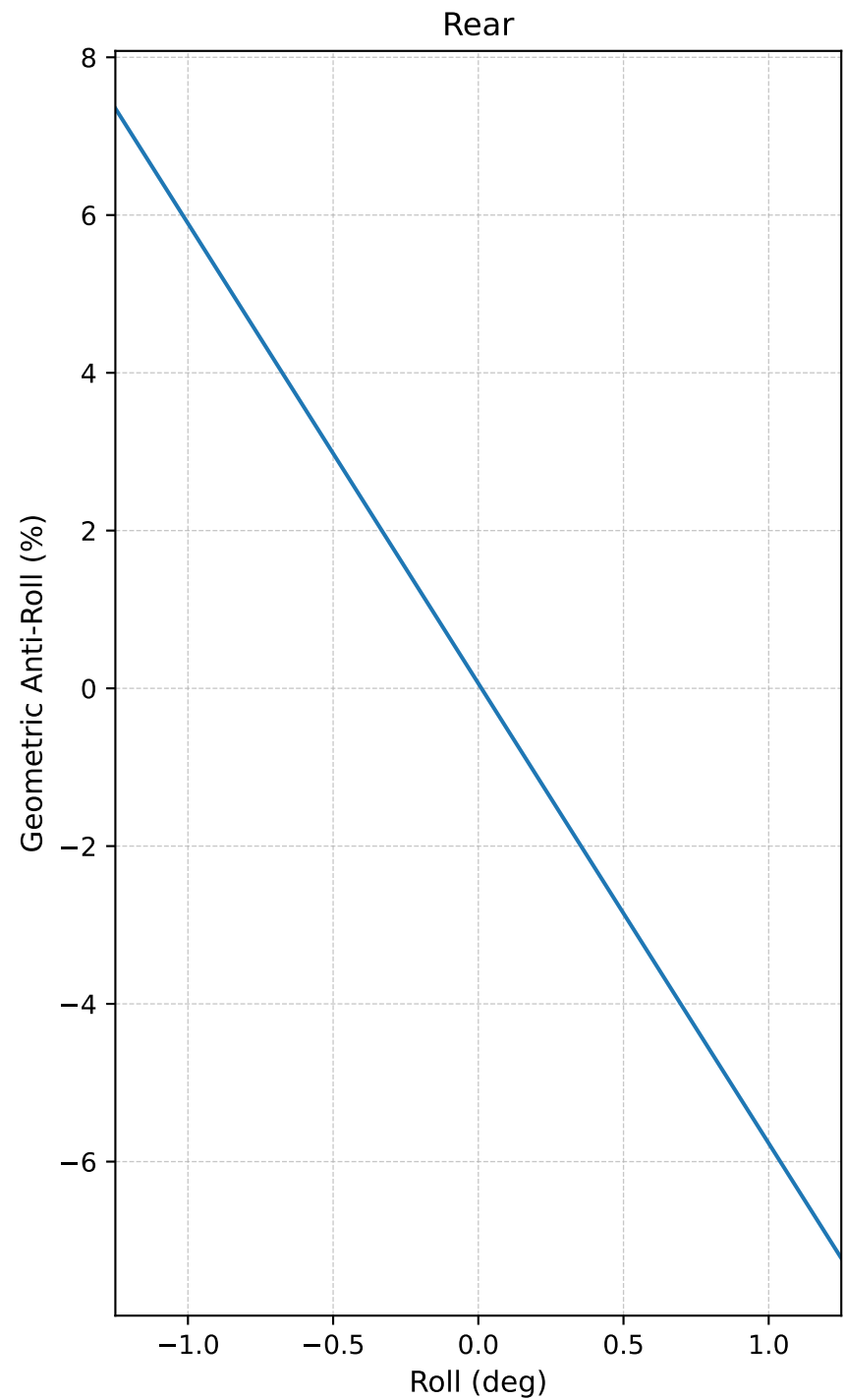
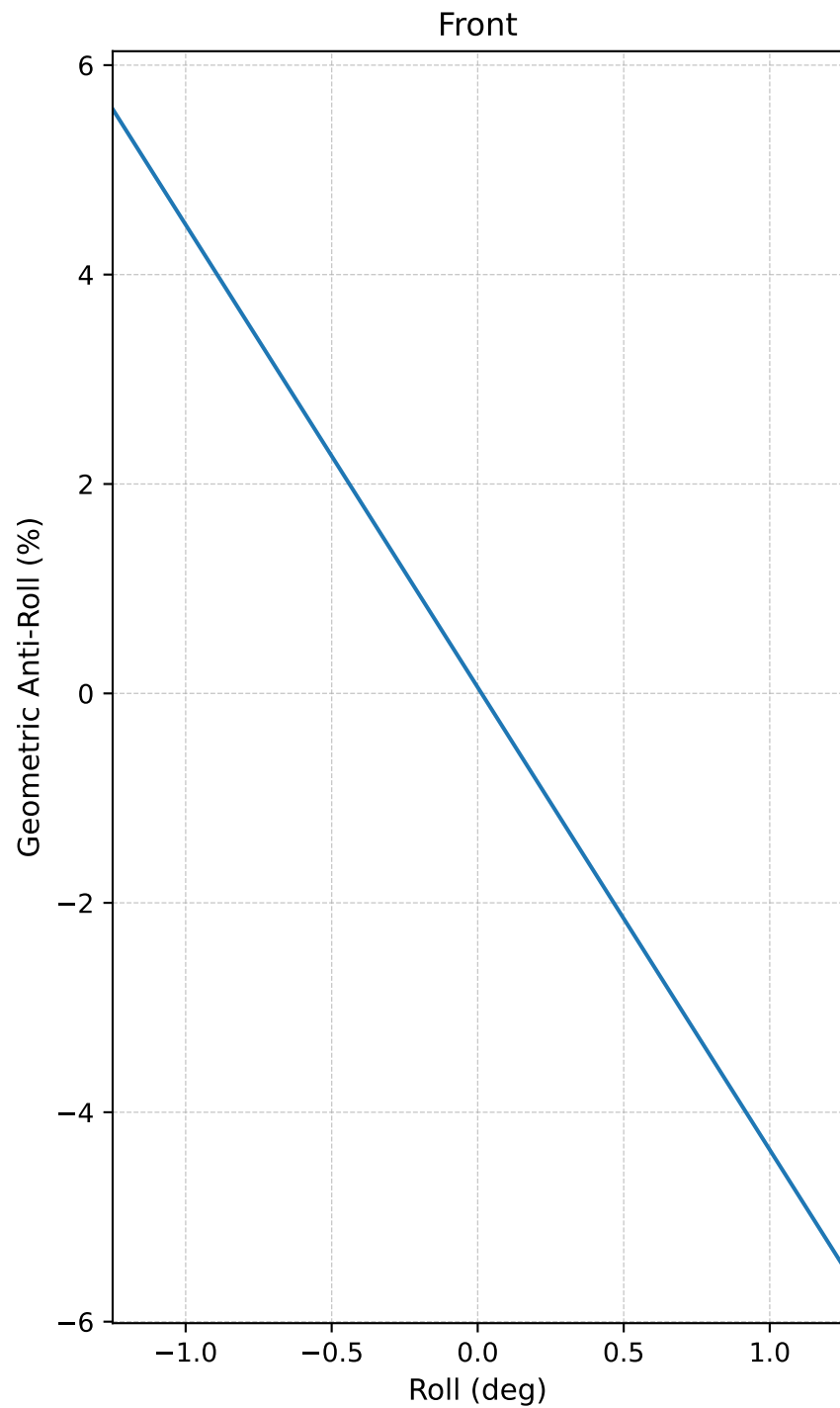
Front



Rear



Geometric Anti-Roll vs Roll



Load-Transfer LLTD vs Roll: Measured vs Roll-Rate Distribution

